

Response regimes in equivalent mechanical model of moderately nonlinear liquid sloshing

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Abstract The paper considers stationary and nonstationary responses in reduced-order model (ROM) of partially liquid-filled tank under external forcing. The model involves one common degree of freedom for the tank and the non-sloshing portion of the liquid, and the other one—for the sloshing portion of the liquid. The coupling between these degrees of freedom is nonlinear, with the lowest-order potential dictated by symmetry considerations. Since the mass of the sloshing liquid in realistic conditions does not exceed 10% of the total mass of the system, the ROM turns to be formally equivalent to well-studied oscillatory systems with nonlinear energy sinks (NES). Exploiting this analogy, and applying the methodology known from the studies of the systems with NES, we predict a multitude of possible nonstationary responses in the considered ROM. These responses conform, at least on the qualitative level, to the responses observed in experimental sloshing settings, multimodal theoretical models and full-scale numeric simulations.

Keywords Liquid sloshing · Partially filled tanks · Reduced-order model · Nonlinear energy sink

List of symbols

m_0	Mass of the ‘static’ liquid portion
m_1	Mass of the first sloshing mode
m_2	Mass of the second sloshing mode
m_3	Mass of the third sloshing mode
m_n	Mass of the n th sloshing mode
m_F	Total fluid mass
h	Undisturbed water level
H	Cylindrical tank height
R	Cylindrical tank radius
$u_g(t)$	Dimensional horizontal ground excitation
l	Rectangular tank breadth
h^*	Critical water level of transition from hardening to softening nonlinearity associated with liquid sloshing
$F(t)$	Harmonic external force
$\bar{A}$	Dimensional forcing amplitude
Ω	Dimensional forcing frequency
t	Dimensional time scale
m	Mass of the internal particle
M	Mass of the primary structure
u	Normalized coordinate related to the primary structure
v	Normalized coordinate related to the internal particle
k_1	Linear stiffness associated with the primary structure
c	Linear viscosity associated with the internal particle

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k_2	Dimensional linear stiffness coefficient associated with the internal particle
$\bar{k}$	Dimensional cubic stiffness coefficient associated with the internal particle
τ	Non-dimensional time scale
ω_1, ω_2	Frequency parameters
σ	Non-dimensional detuning parameter with respect to ω_1
ε	Small parameter
εA	Small non-dimensional forcing amplitude
$\bar{\beta} = \varepsilon \beta$	Small detuned square of the natural frequencies ratio
k	Non-dimensional cubic stiffness coefficient
λ	Non-dimensional linear damping coefficient
X	Non-dimensional coordinate proportional to the displacement of the center-of-mass
w	IP relative displacement with respect to the PM
$\varepsilon \eta$	Small non-dimensional linear stiffness coefficient associated with w
φ_1, φ_2	Complex modulation variables related to X and w , respectively
$\varphi_{10}, \varphi_{20}$	Steady-state values of complex modulation variables φ_1, φ_2
N_0, θ_0	Steady-state amplitude and phase of complex modulation variable φ_2
$Z_0 = N_0^2$	The square of the steady-state amplitude of complex modulation variable φ_2
$\lambda_{\text{SN,cr}}$	Maximum value of λ for saddle-node bifurcation
$A_{\text{SN,cr}}$	Maximum value of A for saddle-node bifurcation
δ_1, δ_2	small perturbation variables of order ε
Ω_{H}	Frequency of modulations due to Hopf bifurcation
$\lambda_{\text{H,cr}}$	Critical damping value for existence of Hopf bifurcation
$T_n = \varepsilon^n \tau$	Non-dimensional time scales
$D_i = \frac{\partial}{\partial T_i}$	Derivation operator with respect to time scale T_i
$C(T_1)$	Arbitrary function of slow time scale T_1
$\Phi(T_1)$	Steady-state value of φ_2 with respect to time scale T_0
N, θ	Amplitude and phase of complex modulation variable $\Phi(T_1)$
$Z = N^2$	The square of the amplitude of complex modulation variable $\Phi(T_1)$

1 Introduction

Partially filled liquid storage tanks of different shapes are used in many fields of engineering, including vehicles [1–3], sea crafts [4], aircrafts [5–7], for the storage of various, maybe hazardous, liquids. The term “sloshing” refers to oscillatory relative motion of the liquid with respect to the containing vessel. The liquid sloshing may be hazardous for the vessel safety, since dynamic loads related to the sloshing may have direct and rather strong calamitous effect on the vessel stability and robustness.

While being most interesting and potentially hazardous, high-amplitude liquid sloshing in partially filled vessels still lacks thorough analytic representation. The sloshing liquid has infinite number of degrees of freedom; boundary conditions on the free surface are nonlinear and time dependent. Near resonance, one can encounter high-amplitude steep sloshing waves and hydraulic jumps. In the former case, nonlinear dynamical features may take place, for example multiple periodic solutions (‘jump’ phenomenon) [5], weakly quasiperiodic [8,9] and strongly modulated responses [10]. Based on experimental results, it was pointed that cubic spring seems to describe the dynamical regimes in the best way [11]. When the hydraulic jumps are involved, significant hydraulic impacts are applied to the vessel inner walls [4]. Hydraulic jumps and wave collisions with the vessel shells express the strong nonlinearities in the system [12–15].

Dynamics of the liquid sloshing in partially filled vessels depends on the liquid depth. For shallow free surface, waves and hydraulic jumps show vicinity of resonance and apply high loads on vessel walls. Lepelletier and Raichlen [16] show that in rectangular tanks under horizontal excitation experimental results are in good agreement with the linear sloshing theory. Forced sloshing and nonlinear resonance phenomena were studied by Moiseev [17]. However, as the excitation frequency approaches the natural frequency of the lowest sloshing mode, nonlinear and dissipative theory must be used. Hydraulic jumps and wave collisions with vessel shells are the source of strong nonlinearities in the system. Moreover, they can cause significant local stresses in the containing structure and can affect the structure stability, resistance and robustness.

Due to computational complexity and cumulative error originated in the complex interaction between different bulks, the numerical solvers have limited ability

to accurately describe the loads caused by the fluid-structure interactions over long time intervals. That is the motivation to look for analytical or semi-analytical methods. Comprehensive studies were made to identify and predict nonlinear sloshing regimes that appear near the resonance, with the help of asymptotic approaches. Most of those methods were based on assumptions of smooth container inner walls, and neglected viscosity [12, 13, 18]. Therefore, potential flow theory can be applied. In this work, we focus on tanks with walls perpendicular to the undisturbed liquid free surface. Even though steady-state asymptotic modal theory might have a limited ability to quantitatively describe sloshing regimes in certain conditions (for example, when water impact on the tank top takes place), it may be instrumental for analysis of the moderately nonlinear responses, their stability and sensitivity to external perturbations. Moreover, the major advantage of the ROM is tiny computational efforts compared to the CFD (computational fluid dynamics) methods. Besides, the coupling between sloshing and the containing structure due to stochastic external excitation can be analyzed for long time intervals, and stability of the response regimes can be analyzed analytically.

Hill [19] and Hill and Frandsen [20] studied the wave height and nonlinear beating amplitude by solving analytically the third-order amplitude evolution equation in terms of Jacobi elliptic functions and considering a single sloshing mode. Their solution is valid for finite depth and non-breaking waves. Steady-state asymptotic modal theory succeeded to predict and describe the observed and documented modulated, or beating sloshing phenomenon [8]. Stolbestov [21] theoretically studied the nonlinear sloshing regimes in almost-square-base container under harmonic excitation along one of the base walls in vicinity of the sloshing lowest natural frequency. His approach was based on the perturbation method introduced by Narimanov [22] that assumed two steady-state sloshing regimes of planar waves and swirling wave. Due to embedded blunder found years later, this method failed to predict more sloshing regimes. As described by Bryant and Stiassnie [23], the following nonlinear regimes might take place in square-base container: planar two-dimensional wave, diagonal wave and swirling wave. They showed that near the primary resonance with respect to the fundamental sloshing mode, the two-dimensional regimes lose their stability for out-of-plane (or three dimensional) perturbation, whereas the three-dimensional regimes

are stable. The same steady-state nonlinear sloshing regimes under longitudinal harmonic excitation were identified by Waterhouse and Ockendon [24, 25] in the vicinity of multimodal nonlinear resonance. Based on Narimanov [22], Faltinsen et al. [18, 26–28] developed multimodal approach to investigate the nonlinear sloshing in a square-base container, using Bateman-Luke variational principle. This method yields multidimensional system of coupled ordinary differential equations of the time dependent multiple natural modes. A weakly nonlinear analysis near resonance of two-dimensional ideal fluid horizontally forced rectangular tank was presented by Forbes [29]. He introduced a novel spectral method which is a generalization of the technique presented by Faltinsen et al. [8]. This analytical method describes both regular and irregular liquid sloshing motion, and managed to obtain irregular solutions that were described by Chester and Bones [30, 31] and numerically by Frandsen [9]. In the work done by Ikeda et al. [32, 33] on nonlinear sloshing responses in square tank under obliquely horizontal excitation, super-position of two modes sloshing motion is combined to yield a swirling motion.

Due to the complex nature of the nonlinear sloshing dynamics and additional difficulty originating from the different tank shapes (i.e., container walls are not vertical and straight), a variety of numerical approaches were developed. Three methods are widely used for sloshing simulations: finite-difference method (FDM, [34]), finite element method (FEM, [35]) and boundary element method (BEM, [36]). The latter is more recommended from reasons of computational complexity and time consumption [10]. Liu and Lin [37] applied the volume of fluid (VOF) method to track free-surface movements under six degrees of freedom containing rectangular tank motion. Their results for horizontal excitation were compared to linear analytical solution given by Faltinsen [38], and three-dimensional violent sloshing simulations were presented. Zhang [39] showed a semi-Lagrangian procedure applied to simulate fully nonlinear sloshing waves in non-wall-sided tank (i.e., tank without vertical walls). His results were compared to open source CFD software OpenFOAM.

Due to the significant computational efforts required for simulation of the complete system, a number of approximate phenomenological models were developed in order to get at least qualitative insight into this phenomenon. A well-known approximate method for sloshing dynamical features imitation and result-

ing hydraulic forces estimation is equivalent mechanical model. Equivalent mechanical models for linear small-amplitude sloshing were based on pendulums, and mass-spring systems were studied for different excitations and tank shapes [5, 6, 40]. However, since nonlinear sloshing regimes cannot be described by the linear models, major attempts were made to formulate nonlinear reduced-order equivalent mechanical models [12, 13, 15, 32, 41, 42] that may predict and describe nonlinear responses observed experimentally. For instance, nonlinear rotary sloshing in a scale model of Centaur propellant tank at low fill level was modeled by combination of ordinary linear pendulum and a spherical pendulum (i.e., compound pendulum) by Kana [43]. In previous works, we treated hydraulic impacts regime with the help of reduced-order mechanical models. Both pendulum [15] and mass-spring systems [44, 45] were used to describe the vibro-impact regimes induced under a horizontal ground excitation. Tank elasticity and strongly nonlinear sloshing were considered, and different dynamical regimes were described with asymptotic tools that were validated numerically. In [45], analysis of realistic partially filled tank was shown and mitigation of vibrations was optimized with the help tuned mass damper (TMD) and NES.

So far, detailed analytical explorations of liquid oscillations in a partially filled vessel under horizontal harmonic ground excitation are limited to small-amplitude sloshing in rectangular and cylindrical vessels. We use the term “moderate” to denote the response regimes, in which the nonlinearity qualitatively modifies the responses, and leads to creation of the new ones, absent in the linear setting. However, the most violent nonlinear sloshing phenomena, such as impacts of the liquid applied to the vessel walls, are excluded. While being interesting and potentially dangerous, the moderate-amplitude liquid sloshing still lacks comprehensive analytic description.

To get (albeit limited) insight into such regimes of the moderately nonlinear sloshing, in the current research we adopt and consider equivalent mechanical model, that attempts to capture the moderate-amplitude sloshing regime. As mentioned above, the moderate-amplitude sloshing exhibits certain dynamical characteristics qualitatively similar to the cubic spring, such as jump phenomenon associated with softening or hardening effects. Besides, the cubic spring is the lowest-order nonlinear term that obeys the symmetry require-

ments. Therefore, in the current study the moderate-amplitude sloshing regime is modeled by addition of a cubic spring to the linear stiffness; thus, the model is relevant both for linear and moderately nonlinear sloshing regimes.

In many engineering partially filled tanks, the lowest modal frequency associated with the tank structure bending is much larger than the fundamental sloshing frequency; it is reasonable to assume that the frequency ratio may be relatively small. Consequently, the effects of the linear and cubic nonlinear stiffness components can become comparable, or the nonlinear stiffness can play the leading role. Moreover, since the mass of the sloshing liquid in realistic conditions does not exceed 10% of the total mass of the system, the ROM turns to be formally equivalent to well-studied oscillatory systems with cubic NES [46–49]. The NES is defined as an essentially nonlinear oscillator with relatively small mass, attached to a primary mechanical system. The model of the sloshing mode adopted in this paper is therefore equivalent to the cubic NES with nonzero linear stiffness component. This addition is crucial for the description of the sloshing motion and puts the considered model somewhat aside the majority of current NES literature. Still, the methods of dynamical analysis developed in the aforementioned papers turn out to be adequate also for the considered model. Conditions for existence and coexistence of periodic steady-state, weak quasiperiodic and strongly modulated responses (SMR) are obtained. The slow invariant manifold (SIM) describing the system slow-flow dynamics of 1:1 internal resonance is derived with the help of multiple-scale analysis of the system. Analytical predictions are validated using numerical simulations. Overview of different analytical, numerical and experimental methods is presented, and the sloshing regimes described by those methods are discussed. Finally, the results are compared qualitatively with previous experimental and computational data and show reasonable qualitative agreement in terms of dynamical regimes.

2 Description of the model

2.1 Introducing the model

Mechanical models of the liquid sloshing induced by ground excitation often use an infinite series of pendula

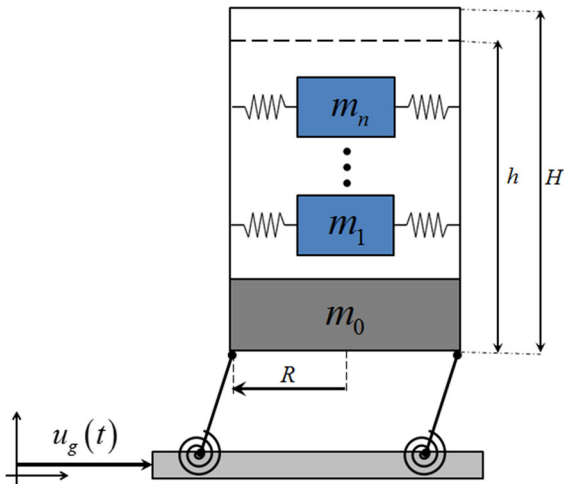


Fig. 1 Scheme of multiple sloshing modes in partially filled tank represented by a series of mass-spring systems interacting with structure walls

or mass-spring systems to represent the free liquid surface oscillation using infinite series of sloshing modes of masses m_n . In Fig. 1, the ground excitation is denoted by $u_g(t)$, undisturbed fluid level by h , and tank radius and height by R and H , respectively. These models, primarily developed using linear sloshing theory, are shown for various types of tank geometries and excitation types, by Graham [50], Graham and Rodriguez [40] and others [5, 6, 11]. Equivalent moment of inertia of a liquid in cylindrical containers has been estimated numerically by Partom [51] and verified experimentally by Werner and Coldwell [52]. Parameters of equivalent pendulum, which corresponds to the first asymmetric sloshing mode of cylindrical and rectangular tanks, were studied by Dodge [6] and Abramson. As shown by Dodge [6], the mass of each modal pendulum decreases rapidly with increasing mode number (Fig. 2). Consequently, it is reasonable to take into account only the first mode in the mechanical equivalent model, as long as the excitation frequency is far from the natural frequencies of the higher modes. The shortcoming of this assumption is that besides disregarding the contribution of higher individual modes, we disregard the possible nonlinear interactions between different sloshing modes that arise near resonance [53].

Extensive use was made for engineering purposes, for example to estimate the hydraulic loads exerted on tanker truck inner walls during breaking and turning and comparison of the results with finite element simu-

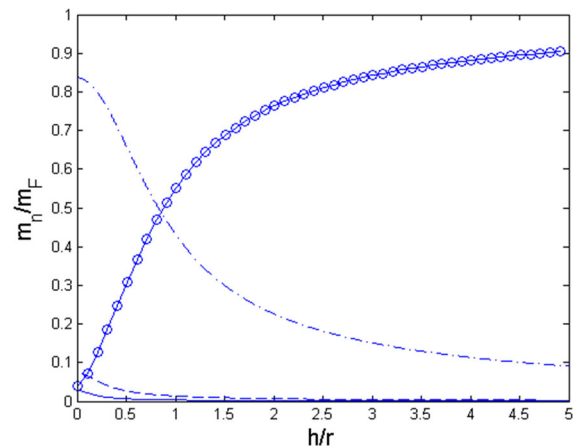


Fig. 2 Ratios of the first three asymmetric sloshing modal masses m_1 , m_2 and m_3 and fixed mass m_0 to the total fluid mass m_F versus the ratio between the liquid depth (h) and the tank radius (R), for cylindrical vessel [44]; solid line with circles: m_0/m_F , dashed dotted line: m_1/m_F , dashed line: m_2/m_F , solid line: m_3/m_F

lation [54], and for resistance prediction of liquid-filled vessels under earthquakes [55].

However, the ROMs based on linear sloshing theory suffer from a major shortcoming. Due to their linearity, they fail to describe nonlinear dynamical regimes, such as hydraulic impacts, rotary sloshing, amplitude jump phenomenon, nonlinear beatings, strongly modulated response, and nonlinear interaction between sloshing and structural modes.

The ROM presented in the current study purports to treat both linear sloshing oscillations, modeled by concentrated mass attached by linear spring to larger containing mass, and cubic nonlinearity representing both linear low-amplitude, and nonlinear high-amplitude sloshing motion. It is, noteworthy, that three-dimensional sloshing regimes of rotary sloshing (i.e., swirling) are not addressed in the discussed model. The reason for this, beyond simplicity considerations, is that we primarily purported to describe analytically the resonant regime, which is approximately two-dimensional, and can be described as such.

Previous analytical studies indicate that response regimes in partially filled tanks subjected to harmonic horizontal excitation are described by Duffing equation, which corresponds to weakly nonlinear behavior of mass-spring system [17]. Moreover, Moiseev showed that even though the asymptotic analysis treated steady-state solutions, comparison with exper-

imental results shows that its predictions are valid also for nonlinear transient waves description. An important conclusion of this analysis was that two-dimensional liquid flow steady-state response changes from hard-spring to soft-spring response with increasing fluid water level. In two-dimensional rectangular tank with breadth l and liquid depth h , this transition occurs for critical normalized water level of $h^*/l \approx 0.3368$. Experiments made by Fultz [56] estimated this value to be $h^*/l = 0.28$. According to Hermann and Timokha [57], this normalized critical water depth decreases monotonously with decreasing ratio of external forcing excitation amplitude to tank breadth, and the value obtained by Moiseev corresponds to the case for which this ratio tends to zero. The results of Hermann and Timokha were consistent with the experiments of Fultz for similar forcing amplitudes. Gu and Sethna [58], Gu et al. [59] have studied the effect of the liquid critical depth in rectangular tanks subjected to vertical sinusoidal excitation.

Following the arguments mentioned above, the model comprises linear oscillator subjected to harmonic external forcing $F(t) = \bar{A} \cos(\Omega t)$, with internal particle with masses of M and m , respectively, where M is the total mass of the container and the non-sloshing liquid portion, and m is the liquid first sloshing mode mass. The internal particle (IP) is attached to the primary structure (PM) by both linear and cubic stiffnesses, where the natural frequency of the smaller mass is much lower than the structure's natural frequency, and for realistic cases of rather filled tanks, its mass is essentially smaller with respect to the mass of the containing structure as can be learned from Fig. 2. Scheme of the system is presented in Fig. 3. Absolute non-dimensional displacements of the primary mass and particle are denoted as $u(t)$ and $v(t)$, respectively.

The PM is modeled by a mass M with linear stiffness k_1 . The IP is located inside a straight cavity in the PM, and in contrast to earlier explored vibro-impact systems [60] and vibro-impact NES [61, 62], it is attached to it through a linear viscosity c and both a linear spring with stiffness k_2 and a cubic spring with stiffness $\bar{k}$. The mass of the first sloshing mode is denoted by m . Values of the modal mass and stiffness of the first sloshing mode, m and k_2 , are obtained using the correlations shown by Abramson [5]. Moreover, the damping coefficient is assessed experimentally for various tank geometrical properties, liquid depth and kinematic viscosity. The cubic stiffness $\bar{k}$ might be determined directly from the

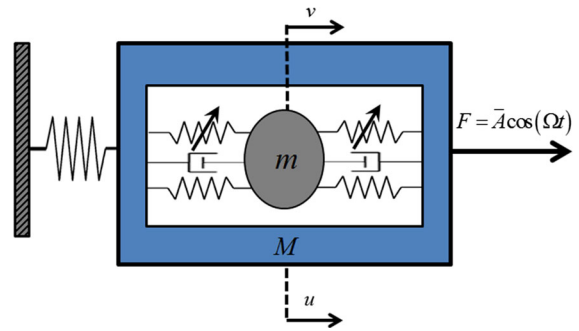


Fig. 3 System scheme-linear oscillator as the primary system and internal particle with both linear and cubic attachments. The displacements of the primary mass and the impacting mass are denoted as u and v , respectively

frequency response curve with the help of experimental tools, as shown by Fultz [56], Hermann and Timokha [57], Gu and Sethna [58] and Gu et al. [59]

The linear spring is required to mimic the small-amplitude oscillatory sloshing motion due to gravity. The additional cubic coupling is required to mimic moderate and high-amplitude sloshing, in accordance with earlier studies [13, 15, 41], which involve nonlinear characteristics associated with a cubic term. Moreover, cubic nonlinearity is the minimal-order symmetric nonlinearity, and therefore, it is the obvious lowest-order correction of the linear coupling term. The external forcing is considered to be harmonic, with frequency Ω and amplitude $\bar{A}$. Hardening nonlinearity corresponds to fluid level of $h < h^*$ and positive value of parameter $\bar{k}$, whereas softening nonlinearity corresponds to $h > h^*$ and negative value of parameter $\bar{k}$.

It is noteworthy that the framework of the presented asymptotic analysis is not limited to any specific tank shape. Any two-dimensional tank shape with vertical walls in the free liquid surface zone can be treated. Moreover, we assume infinite roof height in order to eliminate both liquid spilling and interaction between the fluid and the tank roof.

2.2 Equations of motion

The equations of motion governing the liquid-filled vessel system are formulated with the help of Lagrange formalism. Kinetic and potential energies of the system are expressed as:

$$T = \frac{1}{2} M u_t^2 + \frac{1}{2} m v_t^2$$

$$V = \frac{1}{2}k_1 u^2 + \frac{1}{2}k_2 (v - u)^2 + \frac{1}{4}\bar{k} (v - u)^4 \quad (1)$$

Hence, the Lagrangian is given by the following expression:

$$L = T - V = \frac{1}{2}Mu_t^2 + \frac{1}{2}mv_t^2 - \frac{1}{2}k_1 u^2 - \frac{1}{2}k_2 (v - u)^2 - \frac{1}{4}\bar{k} (v - u)^4 \quad (2)$$

Following Goldstein [63], damping is taken into account through Rayleigh's dissipation function:

$$D = \frac{1}{2}c (v_t - u_t)^2 \quad (3)$$

Thus, by using the extended Hamilton's principle, the dimensional equations of motion are written as follows:

$$\begin{aligned} Mu_{tt} + k_1 u + k_2 (u - v) + c (u_t - v_t) + \bar{k} (u - v)^3 \\ = \bar{A} \cos(\Omega t) \\ mv_{tt} + k_2 (v - u) + c (v_t - u_t) + \bar{k} (v - u)^3 = 0 \end{aligned} \quad (4)$$

We apply simple manipulations and time normalization $\tau = \Omega t$ on Eq. (4) to yield the following normalized non-dimensional equations of motion:

$$\begin{aligned} \ddot{u} + (1 + \varepsilon\sigma)u + \varepsilon^2\beta(u - v) \\ + \varepsilon\lambda(\dot{u} - \dot{v}) + \varepsilon k(u - v)^3 = \varepsilon A \cos(\tau) \\ \ddot{v} + \varepsilon\beta(v - u) + \lambda(\dot{v} - \dot{u}) + k(v - u)^3 = 0 \end{aligned} \quad (5)$$

Here, dots represent differentiation with respect to the normalized time τ , and the normalization and non-dimensional parameters governing the system dynamics are as follows:

$$\begin{aligned} \omega_1^2 = \frac{k_1}{M}, \quad \omega_2^2 = \frac{k_2}{m}, \\ \Omega = \frac{\omega_1}{\sqrt{1 + \varepsilon\sigma}}, \quad \varepsilon A = \frac{\bar{A}}{M\Omega^2} \\ \varepsilon\beta = \left(\frac{\omega_2}{\omega_1}\right)^2 (1 + \varepsilon\sigma), \quad k = \frac{\bar{k}}{m\Omega^2}, \\ \lambda = \frac{c}{m\Omega} \end{aligned} \quad (6)$$

Here, ω_1 and ω_2 are the natural frequencies of the PM and the IM, respectively; $\varepsilon\beta$ is the detuned square of the natural frequencies ratio, εA is a small dimensionless forcing amplitude, k is a non-dimensional cubic stiffness coefficient, and λ is a non-dimensional linear damping coefficient. The following coordinate transformation is used:

$$X(\tau) = u(\tau) + \varepsilon v(\tau)$$

$$w(\tau) = u(\tau) - v(\tau) \quad (7)$$

Here, w the IP relative displacement with respect to the PM, and coordinate X is proportional to the displacement of the center-of-mass. Following [62, 64], the transformed non-dimensional equations of motion:

$$\begin{aligned} \ddot{X} + \frac{1 + \varepsilon\sigma}{1 + \varepsilon}X + \frac{\varepsilon(1 + \varepsilon\sigma)}{1 + \varepsilon}w \\ = \varepsilon A \cos(\tau) \\ \ddot{w} + \frac{1 + \varepsilon\sigma}{1 + \varepsilon}X + \varepsilon\eta w + (1 + \varepsilon)\lambda\dot{w} + k(1 + \varepsilon)w^3 \\ = \varepsilon A \cos(\tau) \end{aligned} \quad (8)$$

Here, $\eta = \frac{1 + \varepsilon\sigma}{1 + \varepsilon} + \beta(1 + \varepsilon)$, $\varepsilon\eta$ is a small non-dimensional linear stiffness coefficient associated with w .

3 Dynamical regimes

As mentioned in previous sections, various nonlinear regimes emerge in partially filled tanks subject to horizontal excitation. In the current section, nonlinear periodic and quasiperiodic sloshing regimes will be described analytically.

3.1 Periodic regimes

The following complex variables are introduced:

$$\begin{aligned} \dot{X} + iX = \varphi_1 e^{i\tau} \\ \dot{w} + iw = \varphi_2 e^{i\tau} \end{aligned} \quad (9)$$

Following Manevitch [65], we rewrite Eq. (8) using the complex transformation shown in Eq. (9) to obtain the following slow-flow system:

$$\begin{aligned} \dot{\varphi}_1 + \frac{i\varepsilon(1 - \sigma)}{2(1 + \varepsilon)}(\varphi_1 + \bar{\varphi}_1 e^{-2i\tau}) \\ - \frac{i\varepsilon(1 + \varepsilon\sigma)}{2(1 + \varepsilon)}(\varphi_2 - \bar{\varphi}_2 e^{-2i\tau}) \\ = \frac{\varepsilon A}{2}(1 - e^{-2i\tau}) \\ \dot{\varphi}_2 + \frac{i}{2}(1 - \varepsilon\eta)(\varphi_2 - \bar{\varphi}_2 e^{-2i\tau}) \\ + \frac{\lambda(1 + \varepsilon)}{2}(\varphi_2 + \bar{\varphi}_2 e^{-2i\tau}) \\ - \frac{i(1 + \varepsilon\sigma)}{2(1 + \varepsilon)}(\varphi_1 - \bar{\varphi}_1 e^{-2i\tau}) \\ + \frac{ik(1 + \varepsilon)}{8}(\varphi_2 e^{i\tau} + \bar{\varphi}_2 e^{-i\tau})^3 e^{-i\tau} \end{aligned}$$

$$= \frac{\varepsilon A}{2} (1 - e^{-2i\tau}) \quad (10)$$

Here, the bar denotes complex conjugate. The problem analyzed is response regime of Eq. (10) in vicinity of both primary and internal resonances of 1:1. Thus, we may assume that the complex modulation variables φ_1 and φ_2 are slow with respect to the external forcing. Thus, by averaging with respect to the fast time scale and neglecting the terms of frequency of 2 and 4, the approximate averaged system is as follows:

$$\begin{aligned} \varphi_1' + \frac{i\varepsilon(1-\sigma)}{2(1+\varepsilon)}\varphi_1 - \frac{i\varepsilon(1+\varepsilon\sigma)}{2(1+\varepsilon)}\varphi_2 &= \frac{\varepsilon A}{2} \\ \varphi_2' + \frac{i}{2}(1-\varepsilon\eta)\varphi_2 + \frac{\lambda(1+\varepsilon)}{2}\varphi_2 \\ - \frac{i(1+\varepsilon\sigma)}{2(1+\varepsilon)}\varphi_1 - \frac{3ki(1+\varepsilon)}{8}|\varphi_2|^2\varphi_2 & \\ = \frac{\varepsilon A}{2} \end{aligned} \quad (11)$$

Differentiation with respect to time is denoted by apostrophe in order to distinguish it from the original fast time scale of the system. The averaging method cannot be considered as rigorous due to its deviation from the averaging theorem validity. However, it was successfully used in many previous studies [42, 66].

Periodic solutions of Eq. (8) correspond to fixed points of Eq. (11). Thus, we eliminate the derivatives to yield the following algebraic equations:

$$\begin{aligned} \frac{i\varepsilon(1-\sigma)}{2(1+\varepsilon)}\varphi_{10} - \frac{i\varepsilon(1+\varepsilon\sigma)}{2(1+\varepsilon)}\varphi_{20} &= \frac{\varepsilon A}{2} \\ \frac{i}{2}(1-\varepsilon\eta)\varphi_{20} + \frac{\lambda(1+\varepsilon)}{2}\varphi_{20} \\ - \frac{i(1+\varepsilon\sigma)}{2(1+\varepsilon)}\varphi_{10} - \frac{3ki(1+\varepsilon)}{8}|\varphi_{20}|^2\varphi_{20} &= \frac{\varepsilon A}{2} \end{aligned} \quad (12)$$

Here, φ_{10} and φ_{20} are the steady-state values of the complex modulation functions, i.e., at the fixed points. Equation (12) can be solved easily to yield the following expression for steady-state solution amplitude:

$$\begin{aligned} N_0^2 \left(a_1^2 + (a_2 + a_3 N_0^2)^2 \right) &= a_4^2 \\ \varphi_{20} &= N_0 e^{i\theta_0} \end{aligned} \quad (13)$$

Expressions for coefficients a_i are given in “Appendix A”. By taking $Z_0 = N_0^2$, we bring Eq. (13) to an equivalent form:

$$\alpha_3 Z_0^3 + \alpha_2 Z_0^2 + \alpha_1 Z_0 + \alpha_4 = 0 \quad (14)$$

Expressions for coefficients α_i are given in “Appendix B”.

Existence of multiple periodic solutions may take place for different parameter values. Transition from qualitatively different periodic solutions corresponds to local bifurcation of the periodic solutions.

3.1.1 Saddle-node bifurcation

In this part, we will find the boundary which separates between single-periodic-solution zone and three-periodic-solution zone in the parameter space. These transitions correspond to saddle-node (SN) bifurcation in Eq. (14). The boundary is obtained by the necessary and complete condition for SN bifurcation of equating the derivative of Eq. (14) to zero:

$$3\alpha_3 Z_0^2 + 2\alpha_2 Z_0 + \alpha_1 = 0 \quad (15)$$

Using polynomials (14) and (15), we obtain the following expression for the SN boundary:

$$\begin{aligned} 3\alpha_3 (\alpha_1 \alpha_2 - 9\alpha_3 \alpha_4)^2 \\ + 2\alpha_2 (\alpha_1 \alpha_2 - 9\alpha_3 \alpha_4) (6\alpha_1 \alpha_3 - 2\alpha_2^2) \\ + \alpha_1 (6\alpha_1 \alpha_3 - 2\alpha_2^2)^2 = 0 \end{aligned} \quad (16)$$

Equation (16) corresponds to nullifying of the discriminants of cubic polynomial (14). For a certain parameter set of ε , σ , k the SN bifurcation, described by Eq. (16), is represented by a contour on plane (A, λ) , as shown in Fig. 4. The single solution zone and three solutions zones correspond to the cases of negative and positive values of the discriminant, respectively.

In Fig. 4, the bounded area corresponds to parameter sets of (A, λ) for which three periodic solutions are possible, whereas the outer area corresponds to the parameter sets that allow a single periodic solution.

We substitute expressions α_i into Eq. (16) and obtain polynomial in A or λ . If we take A as a bifurcation variable and λ as a parameter, we can calculate the critical maximum value of λ for SN bifurcation by eliminating the discriminant of Eq. (16) as a polynomial in A .

$$\lambda_{\text{SN,cr}} = \frac{1}{\sqrt{3}} \left| \frac{\sigma}{1-\sigma} + \varepsilon\beta \right| \quad (17)$$

The maximum value of A is calculated by eliminating the discriminant of Eq. (16) as a polynomial in λ , and taking the higher value obtained:

$$A_{\text{SN,cr}} = \frac{4}{9} \sqrt{\frac{2}{|k(\sigma-1)|}} (\sigma - \varepsilon\beta(\sigma-1))^{3/2} \quad (18)$$

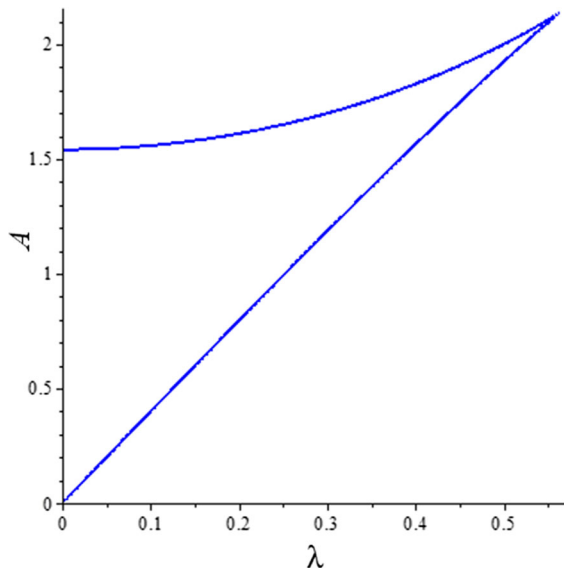


Fig. 4 Projection of the saddle-node bifurcation for $\varepsilon = 0.05$, $k = 4/3$, $\beta = 5$, $\sigma = 5$

As one can see from Eq. (18), necessary condition for existence of SN bifurcation is positive value under the square root. This condition yields the following quadratic polynomial inequality in σ :

$$k(\bar{\beta} - 1)\sigma^2 + k(1 - 2\bar{\beta}) + k\bar{\beta} < 0 \quad (19)$$

Here, $\bar{\beta} = \varepsilon\beta$. The roots of polynomial (19) are as follows:

$$\sigma_{1,2} = 1 + \frac{1 \pm \operatorname{sgn}(k)}{2(\bar{\beta} - 1)} \Rightarrow \sigma_1 = 1, \quad \sigma_2 = \frac{\bar{\beta}}{\bar{\beta} - 1} \quad (20)$$

For positive values of k , SN bifurcations are obtained for $1 < \sigma < \bar{\beta}/(\bar{\beta} - 1)$ if $\bar{\beta} > 1$, and for $\sigma < \bar{\beta}/(\bar{\beta} - 1)$ or $\sigma > 1$ if $0 < \bar{\beta} < 1$. For negative values of k , SN bifurcations are obtained for $\bar{\beta}/(\bar{\beta} - 1) < \sigma < 1$ if $0 < \bar{\beta} < 1$, and for $\sigma < 0$ or $\sigma > \bar{\beta}/(\bar{\beta} - 1)$ if $\bar{\beta} > 1$.

For the special case of $\bar{\beta} = 0$ and $k > 0$, described by Starosvetsky and Gendelman [49], SN bifurcations are obtained for $\sigma < 0$ or $\sigma > 1$. Isocline maps of $A_{\text{SN,cr}}$ and $\lambda_{\text{SN,cr}}$ versus σ and $\bar{\beta}$ for $k = 4/3$ are shown in Fig. 5a, b, respectively.

3.2 Quasiperiodic regimes

In the following section, we refer to two types of quasiperiodic responses; the first one is weak quasiperi-

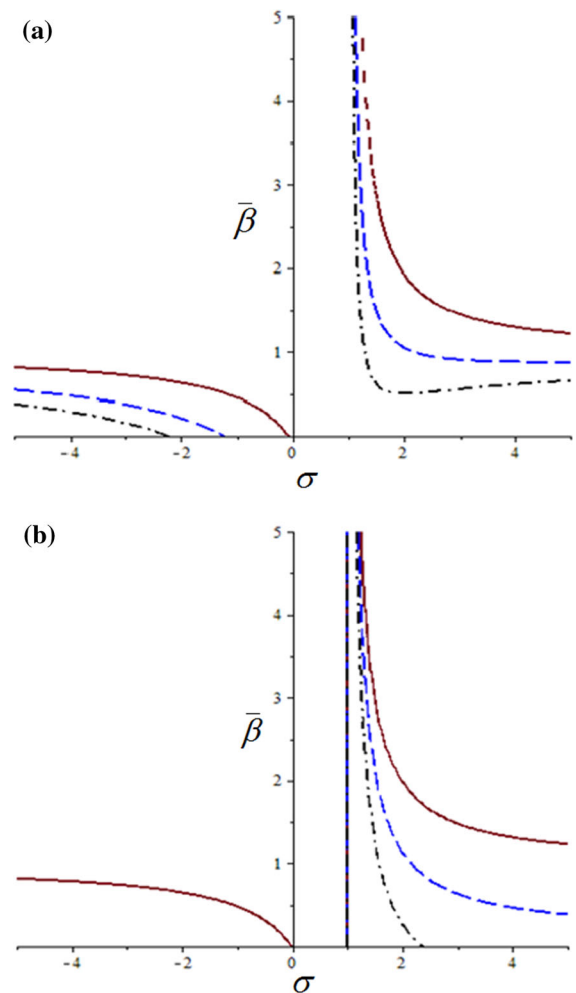


Fig. 5 Isocline maps of A_{cr} (a) and λ_{cr} (b) for existence of SN bifurcation versus σ and $\bar{\beta}$ for $k = 4/3$; **a** solid red: $A_{\text{SN,cr}} = 0.01$, dashed blue: $A_{\text{SN,cr}} = 0.5$, dashed dot: $A_{\text{SN,cr}} = 1$. **b** Solid red: $\lambda_{\text{SN,cr}} = 0.01$, dashed blue: $\lambda_{\text{SN,cr}} = 0.2$, dashed dot: $\lambda_{\text{SN,cr}} = 0.5$. (Color figure online)

odic response, which corresponds to loss of stability of the periodic regime and formation of the limit cycle through Hopf bifurcation. The second quasiperiodic regime is associated with relaxation oscillations, which are characterized by alternating fast and slow response amplitude variations. Detailed investigation of those regimes is described by Starosvetsky and Gendelman [47].

3.2.1 Weak quasiperiodic responses

In order to obtain the conditions for Hopf bifurcation, we perform small perturbation around the fixed points of Eq. (11):

$$\begin{aligned}\varphi_1 &= \varphi_{10} + \delta_1 \\ \varphi_2 &= \varphi_{20} + \delta_2\end{aligned}\quad (21)$$

Here, δ_1 and δ_2 are small perturbation variables, of order ε . We substitute Eq. (21) into the averaged Eq. (11) and keep merely the linear terms to yield the following linearized system, shown in its matrix form:

$$\begin{pmatrix} \delta_1' \\ \delta_1' \\ \delta_2' \\ \delta_2' \end{pmatrix} = \begin{pmatrix} -c_{11} & 0 & c_{12} & 0 \\ 0 & c_{11} & 0 & 0 \\ c_{21} & 0 & -(c_{22} + c_{23}) + 2c_{24}|\varphi_{20}|^2 & 0 \\ 0 & -c_{21} & -c_{24}\bar{\varphi}_{20}^2 & -(c_{22} - c_{23}) - 2c_{24}|\varphi_{20}|^2 \end{pmatrix} \begin{pmatrix} \delta_1 \\ \delta_1 \\ \delta_2 \\ \delta_2 \end{pmatrix} \quad (22)$$

Expressions for coefficients c_{ij} are given in “Appendix C”. The corresponding characteristic polynomial of Eq. (22) is as follows:

$$P(\mu) = \mu^4 + \gamma_1\mu^3 + \gamma_2\mu^2 + \gamma_3\mu + \gamma_4 \quad (23)$$

Expressions for coefficients γ_i are given in “Appendix D”. Hopf bifurcation corresponds to $\mu = \pm i\Omega_H$, where Ω_H is real and refers to the modulation frequency. That implies $\Omega_H^2 = \gamma_1/\gamma_3$ and $\gamma_3^2/\gamma_1^2 - \gamma_2\gamma_3/\gamma_1 + \gamma_4 = 0 \Rightarrow \Delta_3 = 0$. Thus, the condition for Hopf bifurcation and Hopf frequencies is as follows:

$$\begin{aligned}\gamma_3^2 - \gamma_1\gamma_2\gamma_3 + \gamma_4\gamma_1^2 &= 0 \\ \Omega_H &= \pm \frac{1}{2} \sqrt{\frac{\varepsilon(1 + \varepsilon\sigma)}{1 + \varepsilon}}\end{aligned}\quad (24)$$

Locus of SN and Hopf bifurcations on (λ, A) plane for several values of β is shown in Fig. 6.

As one can see in Fig. 6b, that so far the bifurcations locus consists of three different zones: point 1: existence of a single periodic solution; point 2: coexistence two periodic regimes; and point 3: coexistence of both three periodic and weakly quasiperiodic regimes.

Equation (24) is a biquadratic polynomial with respect to N and has two roots. Critical damping value for existence of Hopf bifurcation $\lambda_{H,cr}$ is obtained by equating both of them:

$$\lambda_{H,cr} = \frac{1}{\sqrt{3}} \left| \bar{\beta} + \frac{\varepsilon\sigma - 1}{(1 + \varepsilon)} \right| \quad (25)$$

Isocline maps of $\lambda_{H,cr}$ versus σ and $\bar{\beta}$ for $\varepsilon = 0.05$ are shown in Fig. 7.

As one can see in Fig. 6, each projection of bifurcation manifold of $\beta = 5$ consists of three different zones, corresponding to points 1, 2 and 3. Hopf and SN bifurcations diagrams for fixed values of parameters corresponding to the three points are presented in Fig. 8a–c.

3.2.2 Strongly modulated responses (SMR)

The regions of stability of periodic solutions discussed previously were analyzed by tools of local bifurcations and perturbation methods. Those methods cannot be used for description of the additional existing regime of SMR. As shown in previous studies [47, 49], this regime

is characterized by aggressive amplitude modulations and exists in vicinity of 1:1 resonance. In contrary to the current study, those previous studies focused on essentially nonlinear attachment, known as cubic NES, and its contribution in terms of mitigation performances of linear oscillator. In this section, following Starosvetsky and Gendelman [49], we obtain the conditions for existence of SMR using analytical approach of multiple-scales.

By differentiation and simple algebraic procedures, we reduce Eq. (11) to the following single second-order equations with respect to φ_2 :

$$\begin{aligned}\frac{d^2}{d\tau^2}\varphi_2 + \frac{d}{d\tau} \left(\bar{c}_1\varphi_2 + \bar{c}_2|\varphi_2|^2\varphi_2 \right) \\ - \left(\bar{c}_3\varphi_2 + \bar{c}_4|\varphi_2|^2\varphi_2 + \frac{\varepsilon A}{2}\bar{c}_5 \right) = 0\end{aligned}\quad (26)$$

Expressions for coefficients $\bar{c}_i$ are given in “Appendix E”. Multiple-scales approach is introduced:

$$\begin{aligned}\varphi_2 &= \varphi_2(T_0, T_1) \\ T_n &= \varepsilon^n \tau, \quad n = 0, 1, \dots \\ \frac{d}{d\tau} &= D_0 + \varepsilon D_1\end{aligned}\quad (27)$$

Here, ε is a small parameter and $D_i = \frac{\partial}{\partial T_i}$, $D_i^2 = \frac{\partial^2}{\partial T_i^2}$. Substituting Eq. (27) into Eq. (26) and collecting equal orders terms with respect to small parameter ε obtains the following expressions:

$$\begin{aligned}\mathbf{O}(1) : D_0^2\varphi_2 + D_0 \left[\left(\frac{\lambda}{2} + \frac{i}{2} \right) \varphi_2 - \frac{3ki}{8} |\varphi_2|^2 \varphi_2 \right] &= 0 \\ \mathbf{O}(\varepsilon) : 2D_0D_1\varphi_2 \\ + D_0 \left[\left(\frac{i(1 - \sigma)}{2} + \frac{\lambda}{2} - \frac{i}{2}(1 + \beta) \right) \varphi_2 \right. \\ \left. - \frac{3ki}{8} |\varphi_2|^2 \varphi_2 \right] &\end{aligned}$$

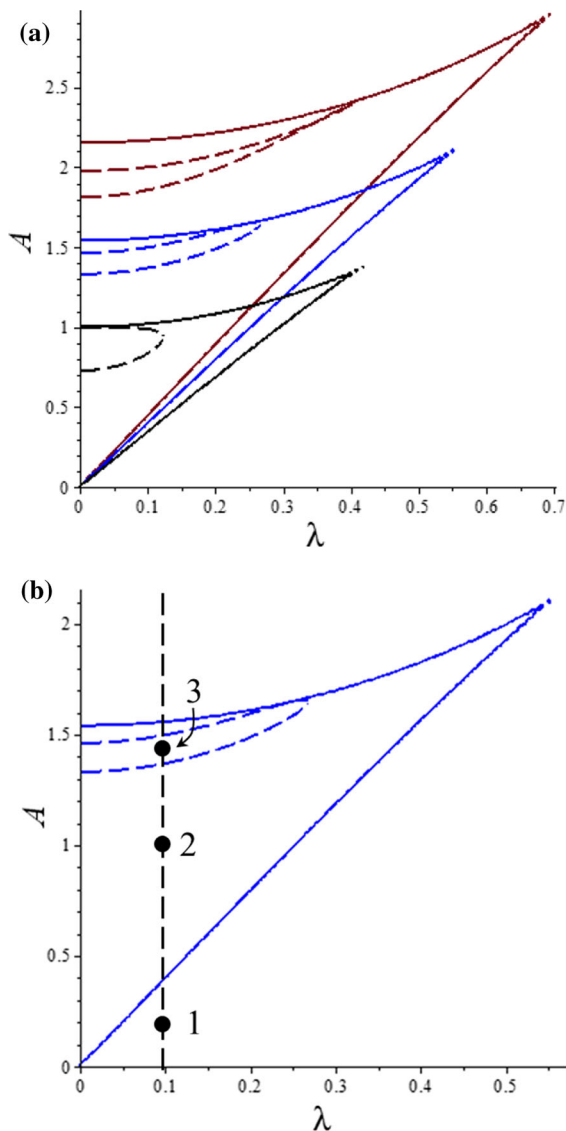


Fig. 6 Projection of the saddle-node and Hopf bifurcations for: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$; SN bifurcations: solid line, Hopf bifurcation: dashed line. **a** Red: $\beta = 0$, blue: $\beta = 5$, black: $\beta = 10$ (color online); **b** for $\beta = 5$, point 1: $A = 0.1$ existence of a single periodic solution; point 2: $A = 1$ coexistence two periodic regimes; point 3: $A = 1.4$ coexistence of both three periodic and weakly quasiperiodic regimes

$$\begin{aligned}
 & + D_1 \left[\left(\frac{\lambda}{2} + \frac{i}{2} \right) \varphi_2 - \frac{3ki}{8} |\varphi_2|^2 \varphi_2 \right] \\
 & - \left(-\frac{1}{4} (1 + i (1 - \sigma) (\lambda + i)) \varphi_2 \right. \\
 & \left. - \frac{3k(1 - \sigma)}{16} |\varphi_2|^2 \varphi_2 + \frac{Ai}{4} \right) = 0
 \end{aligned} \quad (28)$$

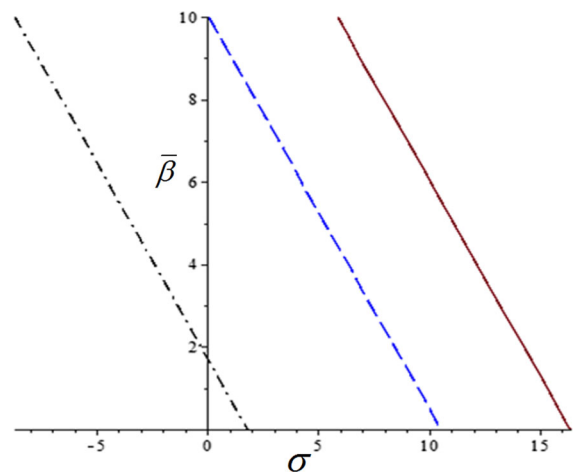


Fig. 7 Isocline maps of $\lambda_{H,cr}$ versus σ and $\bar{\beta}$ for $\varepsilon = 0.05$. Solid red: $\lambda_{H,cr} = 0.1$, dashed blue: $\lambda_{H,cr} = 0.268$, dash dotted black: $\lambda_{H,cr} = 0.5$. (Color figure online)

The solution of the first equation of system (28) represents the fast evolution of the averaged system. Integration of this equation with respect to time scale T_0 yields the following:

$$D_0 \varphi_2 + \left[\left(\frac{\lambda}{2} + \frac{i}{2} \right) \varphi_2 - \frac{3ki}{8} |\varphi_2|^2 \varphi_2 \right] = C(T_1) \quad (29)$$

Here, $C(T_1)$ is an arbitrary function of slow time scale T_1 , i.e., constant with respect to time scale T_0 . Stationary points $\Phi(T_1)$ of Eq. (29) correspond to elimination of the derivative with respect to time scale T_0 , and obtained by the following algebraic equation:

$$\left[\left(\frac{\lambda}{2} + \frac{i}{2} \right) \Phi - \frac{3ki}{8} |\Phi|^2 \Phi \right] = C(T_1) \quad (30)$$

We substitute the polar transformation $\Phi(T_1) = N(T_1) e^{i\theta(T_1)}$ into Eq. (30) and use several simple manipulations:

$$Z \left(\lambda^2 + \left(1 - \frac{3k}{4} Z \right)^2 \right) = 4 |C(T_1)|^2 \quad (31)$$

Here, $Z(T_1) = N(T_1)^2$. As one can see, Eq. (31) may have either a single solution or three solutions. The former case corresponds to monotony of the function in the left-hand side of Eq. (31), and the latter corresponds to existence of extremal points. In order to discern between the different cases, we equate the derivative of Eq. (31), i.e., the condition for existence of the extremum:

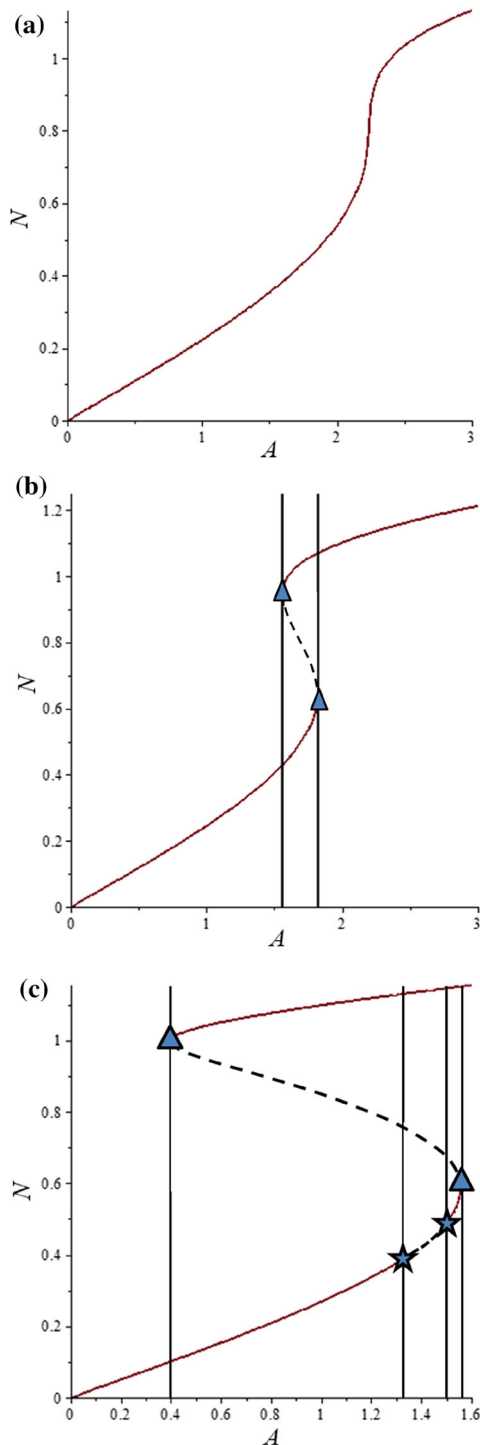


Fig. 8 SN bifurcations (marked by triangles) and Hopf bifurcations (marked by stars) diagram for: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $\beta = 5 \Rightarrow \lambda_{H,cr} = 0.268$, $\lambda_{SN,cr} = 0.577$. Stable solutions—solid lines, unstable solutions—dashed line; **a** $\lambda = 0.6 > \lambda_{SN,cr}$; **b** $\lambda_{H,cr} < \lambda = 0.4 < \lambda_{SN,cr}$ and **c** $\lambda = 0.1 < \lambda_{H,cr}$

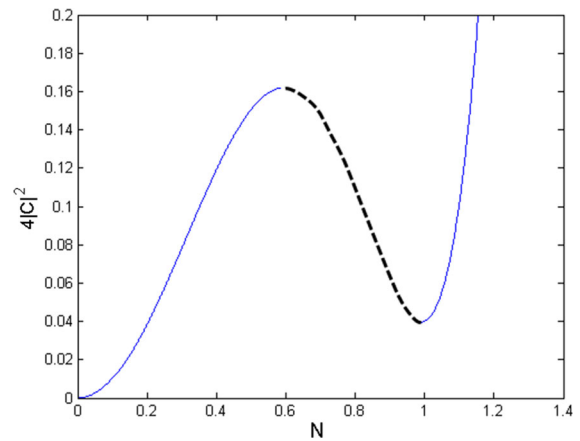


Fig. 9 Projection of the SIM on plane $(N, 4|C(T_1)|^2)$ for: $k = 4/3$, $\lambda = 0.2$

$$\begin{aligned} \frac{27k^2}{16}Z^2 - 3kZ + \lambda^2 + 1 &= 0 \text{ or } Z_{1,2} \\ &= \frac{8 \pm 4\sqrt{1 - 3\lambda^2}}{9k} \end{aligned} \quad (32)$$

The boundary value between single and triplet solutions zones corresponds to $\lambda = 1/\sqrt{3}$ and is referred to as SN bifurcation. It can be easily concluded from Eq. (32), SN bifurcation can take place only for positive value of parameter k , for which polynomial (32) has positive roots, corresponding to real values of $N_{1,2} = \sqrt{Z_{1,2}}$. Thus, relaxation motion exists only for hardening cubic nonlinearity and corresponds to positive value of parameter k .

As a meter of fact, Eq. (31) defines the slow invariant manifold (SIM) of the problem. Projection of the SIM on plane $(N, 4|C(T_1)|^2)$ is presented in Fig. 9. The SN bifurcation points satisfy the second equation of system (32). As shown in previous studies [67], three-solutions zone in the SIM may allow relaxation oscillations of the slow system, represented by fast jumps from one stable branch to the other.

Description of the system motion on the SIM stable branches $\Phi(T_1) = \lim_{T_0 \rightarrow \infty} \varphi_2(T_0, T_1)$ is given by solving the ε order equations of Eq. (28) corresponding to steady-state with respect to time scale T_0 :

$$\begin{aligned} D_1 \left[\left(\frac{\lambda}{2} + \frac{i}{2} \right) \Phi - \frac{3ki}{8} |\Phi|^2 \Phi \right] \\ - \left(-\frac{1}{4} (1 + i(1 - \sigma)(\lambda + i)) \Phi \right. \\ \left. - \frac{3k(1 - \sigma)}{16} |\Phi|^2 \Phi + \frac{Ai}{4} \right) = 0 \end{aligned} \quad (33)$$

After simple manipulations, we get the following:

$$\left(\frac{\lambda + i}{2} - \frac{3ki|\Phi|^2}{4} \right) D_1 \Phi - \frac{3ki}{8} \Phi^2 D_1 \bar{\Phi} = G(\Phi, \bar{\Phi}) \quad (34)$$

After taking the complex conjugate of Eq. (34), we get the following derivative:

$$D_1 \Phi = \frac{(h_1 - ih_2)G + h_3i\Phi^2\bar{G}}{h_1^2 + h_2^2 - h_3^2|\Phi|^4}$$

$$h_1 = \frac{\lambda}{2}, h_2 = \frac{1}{2} - \frac{3k|\Phi|^2}{4},$$

$$h_3 = \frac{3k}{8}G(\Phi, \bar{\Phi}) = \frac{1}{4}(- (1 + i(1 - \sigma)(\lambda + i))\Phi - \frac{3k(1 - \sigma)}{4}|\Phi|^2 + iA) \quad (35)$$

By dividing variable Φ to modulus and argument by using polar transformations $\Phi(T_1) = N(T_1)e^{i\theta(T_1)}$, we get the following averaged slow-flow equations:

$$\frac{\partial N}{\partial T_1} = \frac{-\frac{3k}{4}A \cos \theta N^2 - \lambda N + \lambda A \sin \theta + A \cos \theta}{2(\lambda^2 + 1 - 3kN^2 + \frac{27k^2}{16}N^4)}$$

$$\frac{\partial \theta}{\partial T_1} = \frac{-\frac{27k^2}{16}(1 - \sigma)N^4 + \frac{3k}{4}(1 - 4\sigma)N^2 + \frac{9k}{4}A \sin \theta N + (\sigma - \lambda^2(1 - \sigma)) + \frac{\lambda A \cos \theta - A \sin \theta}{N}}{2(\lambda^2 + 1 - 3kN^2 + \frac{27k^2}{16}N^4)} \quad (36)$$

We denote the numerator of both equations in Eq. (36) as $f_1(N, \theta)$ and $f_2(N, \theta)$, respectively, and their identical denominator as $g(N)$ to yield the following:

$$\frac{\partial N}{\partial T_1} = \frac{f_1(N, \theta)}{g(N)}$$

$$\frac{\partial \theta}{\partial T_1} = \frac{f_2(N, \theta)}{g(N)} \quad (37)$$

From observing Eq. (37), one can distinguish between two different types of fixed points of the slow-flow system. The first one is referred to as regular fixed points of the slow flow and fulfills both: $\partial N / \partial T_1 = \partial \theta / \partial T_1 = 0$ and $g(N) \neq 0$. The points satisfying these conditions fabricate the slow invariant manifold (SIM) of the system. The second type is referred to as folded singularities and fulfills both: $\partial N / \partial T_1 = \partial \theta / \partial T_1 = 0$ and $g(N) = 0$.

Nullifying both denominators $f_i(N, \theta)$ yields the following system of equations:

$$\begin{pmatrix} \xi_{11} & \xi_{12} \\ \xi_{21} & \xi_{22} \end{pmatrix} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix} \quad (38)$$

Expressions for coefficients ξ_{ij} are given in ‘‘Appendix F’’. The ordinary fixed points are calculated from Eq. (38) by multiplying by the inverse of matrix ξ , since its determinant is $\det(\xi) = \xi_{11}\xi_{22} - \xi_{12}\xi_{21} = -\frac{A^2}{2}g(N) \neq 0$:

$$\alpha_3 N_0^6 + \alpha_2 N_0^4 + \alpha_1 N_0^2 + \alpha_4 = 0 \quad (39)$$

Here, α_i are, not surprisingly, are the same as those shown in Eq. (14), which describes the fixed points of the system in terms of $Z_0 = N_0^2$.

The second type of fixed points, additionally to nullifying $f_i(N, \theta)$, satisfies the following condition:

$$g(N) = 0 \Rightarrow \frac{27k^2}{16}N_0^4 - 3kN_0^2 + \lambda^2 + 1 = 0 \quad (40)$$

By solving the biquadratic polynomial (40), one obtains the following solution:

$$N_0 = \sqrt{\frac{8 \pm 4\sqrt{1 - 3\lambda^2}}{9k}} \quad (41)$$

From Eq. (38), we yield the following equations:

$$A \left(1 - \frac{3k}{4}N_0^2 \right) \cos \theta_0 + \lambda A \sin \theta_0 = \lambda N_0$$

$$\lambda A \cos \theta_0 - A \left(1 - \frac{9k}{4}N_0^2 \right) \sin \theta_0$$

$$= \frac{27k^2}{16}N_0^2 - \frac{3k}{4}(1 - 4\sigma)N_0^3 - (\sigma - \lambda^2(1 - \sigma))N_0 \quad (42)$$

Since this type of solutions corresponds to elimination of the discriminant of Eq. (38), both equations included are linearly dependent. Thus, one can treat merely the first one, and after simple trigonometric manipulations, we yield the following expressions:

$$\cos(\theta_0 + \gamma_0) = \frac{\lambda N_0}{\sqrt{(1 - \frac{3k}{4}N_0^2)^2 + \lambda^2}}$$

$$\sin \gamma_0 = \frac{\lambda}{\sqrt{(1 - \frac{3k}{4}N_0^2)^2 + \lambda^2}} \quad (43)$$

Both equations of Eq. (43) are solvable only if their right-hand sides' absolute values do not exceed unity. By equating to unity and using Eq. (41), we obtain the critical excitation amplitude required in order to allow the existence of the regime discussed:

$$A_{\text{SMR},1,2} = \lambda \sqrt{\frac{2}{|k|}} \sqrt{\frac{2 \pm \sqrt{1 - 3\lambda^2}}{1 + 3\lambda^2 \mp \sqrt{1 - 3\lambda^2}}} \quad (44)$$

Obviously, the minimal excitation amplitude for existence of SMR regime corresponds to $A_{\text{SMR},\text{cr}} = \min \{A_{\text{SMR},1,2}\}$:

$$A_{\text{SMR},\text{cr}} = \lambda \sqrt{\frac{2}{|k|}} \sqrt{\frac{2 - \sqrt{1 - 3\lambda^2}}{1 + 3\lambda^2 + \sqrt{1 - 3\lambda^2}}} \quad (45)$$

Hence, in order to observe SMR regime, the following condition should hold: $A > A_{\text{SMR},\text{cr}}(\lambda, k)$.

4 Numerical results

In the following section, we compare the analytical predictions of the modulation envelope of the averaged system with numerical simulations of the full equations of motion shown in Eq. (8). The goals of this section are to validate the asymptotic results obtained in the previous chapter, and to demonstrate the dynamical regimes predicted for different parameter sets. Following Figs. 6b and 8, we analyze the different dynamical regimes by varying the values of parameters A and λ . The corresponding parameter set is as follows: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $\beta = 5$.

The case of single periodic solution, corresponding to Fig. 8a, is shown in Fig. 10 in terms of time history simulation and slow-flow evolution on the SIM.

Appropriate choice of parameters is of special importance for this simulation. The nonlinear stiffness parameter k is rescalable; therefore, its exact value bears no physical meaning. The detuning parameter σ and damping coefficient λ were chosen according to Starosvetsky and Gendelman [49]. The former parameter is related to the external forcing and can be adjusted according to the experimental conditions; the latter should be perceived as empiric coefficient. The mass ratio parameter ε was chosen accordingly to well-known analytical studies made by Abramson [5] and Dodge [6] and dimensional analysis, similar to the one shown in [45]. The linear stiffness coefficient β was chosen phenomenologically. It should be mentioned

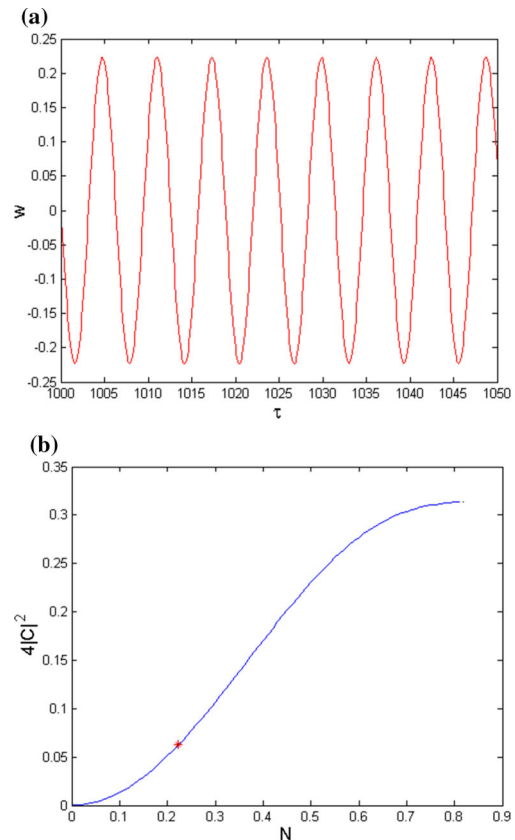


Fig. 10 Single periodic solution, corresponding to Fig. 8a: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $A = 1$, $\beta = 5 \Rightarrow \lambda_{\text{H},\text{cr}} = 0.268$, $\lambda_{\text{SN},\text{cr}} = 0.577$, $\lambda = 0.6 > \lambda_{\text{SN},\text{cr}}$. Initial conditions: $u_0 = v_0 = \dot{u}_0 = \dot{v}_0 = 0$

that the exact parameter values of the equivalent model would strongly depend on geometry of the vessel and other particularities. In general, they can be identified experimentally or with the help of full-scale numerics.

As one can see, periodic solution shown in Fig. 10a is transformed to a single dot on the SIM curve shown in Fig. 10b. Since the SIM curve is monotonous, multiple solutions are not possible.

The case of multiple periodic solution (amplitude ‘jump’ phenomenon), corresponding to Fig. 8b, is shown in Fig. 11, where for $\lambda = 0.4$ and certain range of excitation amplitude A (corresponding to the second region from the left in Fig. 8b), two responses with different amplitudes may be obtained for different initial condition sets.

Similarly to Fig. 10, two periodic responses with different amplitudes are transformed to two different dots on the SIM curve. Multiple solutions are possible due to

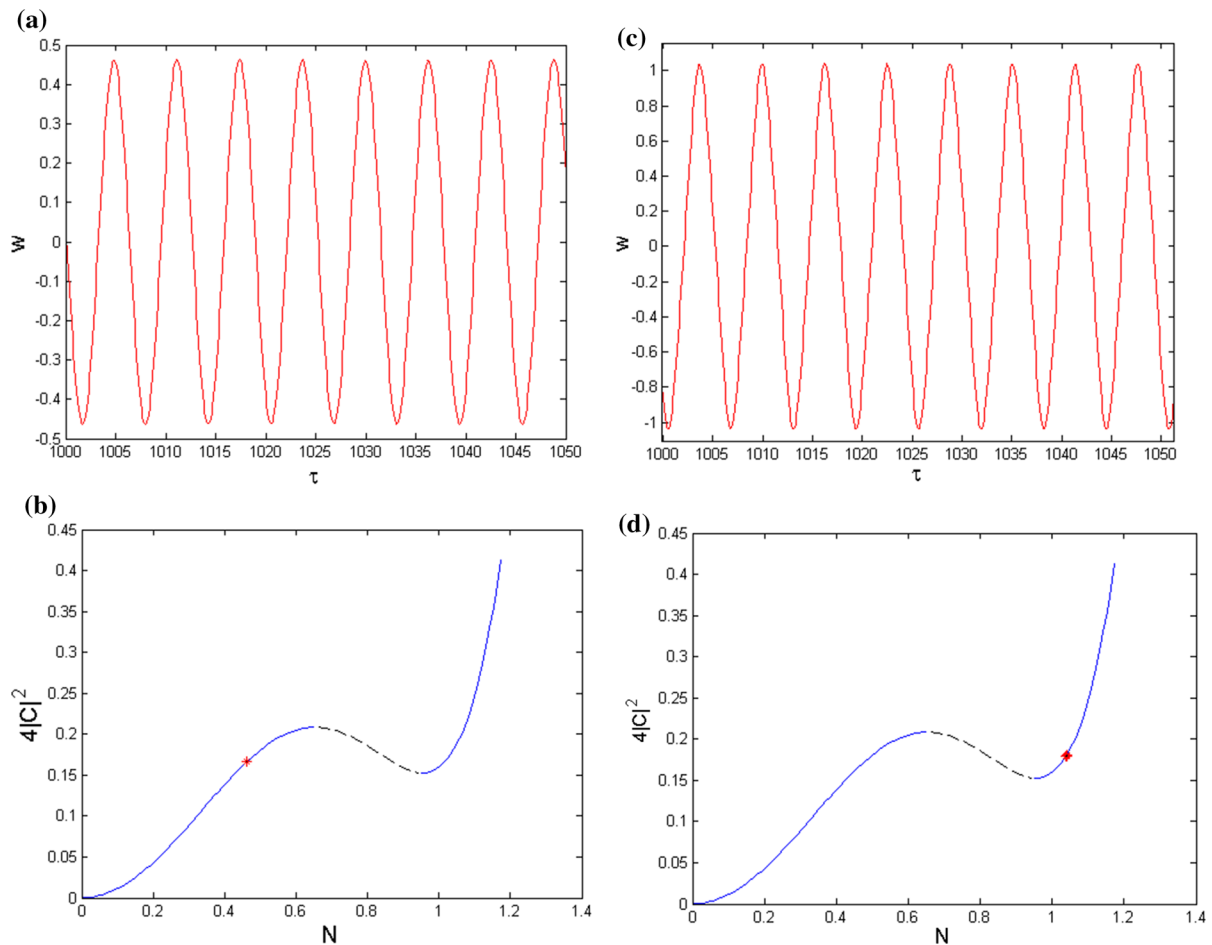


Fig. 11 Double periodic solution, corresponding to Fig. 8b: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $A = 1.62$, $\beta = 5 \Rightarrow \lambda_{H,cr} = 0.268$, $\lambda_{SN,cr} = 0.577$, $\lambda_{H,cr} < \lambda = 0.4 < \lambda_{SN,cr}$. Initial conditions: **a** $u_0 = v_0 = \dot{u}_0 = \dot{v}_0 = 0$ and **b** $u_0 = v_0 = 0, \dot{u}_0 = 0.5\dot{v}_0 = 0$

the bending of the SIM curve shown in Fig. 11b, in contrast to the monotone SIM curve shown in Fig. 10b. This coexistence of both periodic solution is associated with the jump phenomenon, since the system's slow flow, i.e., steady-state amplitude, may either settle on the left (smaller amplitude) branch, or on the right branch (higher amplitude).

The case of coexistence of both periodic solution and weakly quasiperiodic regime, corresponding to Fig. 8c, is shown in Fig. 12, where for $\lambda = 0.6$ and a certain range of excitation amplitude A (corresponding to the third region from the left in Fig. 8c) two responses with different amplitude are obtained for different initial condition sets.

The slow evolution of response amplitude (weak modulations) shown in Fig. 12a is transformed to 'traveling' dots on the SIM stable branch, as shown in

Fig. 12b. As one can see in Fig. 12b, the slow flow approaches the point of contact between the SIM stable and unstable branch, corresponding to local maximum of the curve near $N = 0.6$. However, since the slow flow does not manage to reach the local maximum, it remains on the left branch and does not jump to the right branch. Hence, merely weak modulations take place.

Finally, the case of coexistence of both periodic and strongly quasiperiodic regimes, corresponding to the case in which $A > A_{SMR,cr}$, is shown in Fig. 13.

To guide the eye, sequential dots are connected by a solid black line. Now, the system's slow flow reaches and crosses the point of contact and immersed to the right stable branch, causing a sharp increase in the response amplitude envelope. The sharp increase

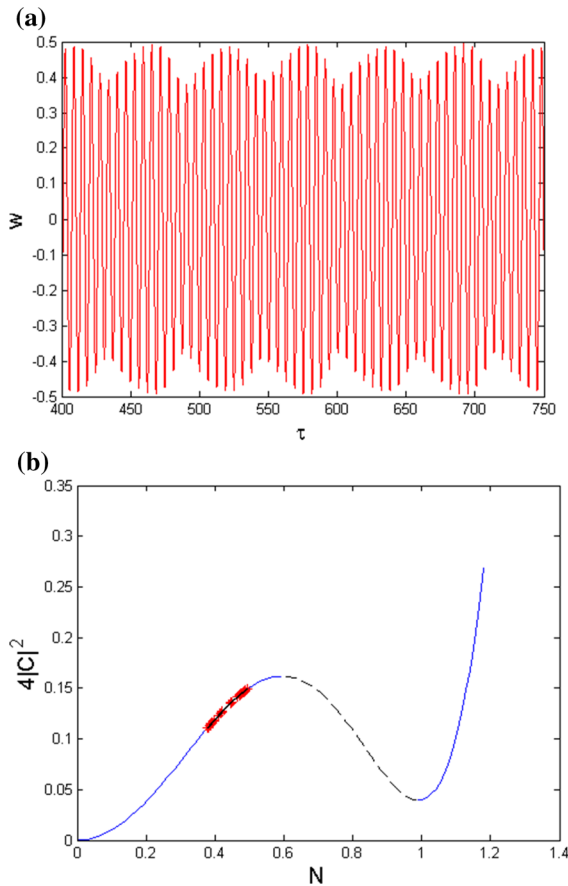


Fig. 12 Quasiperiodic solution, corresponding to Fig. 8c: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $A = 1.4$, $\beta = 5 \Rightarrow \lambda_{H,cr} = 0.268$, $\lambda_{SN,cr} = 0.577$, $\lambda = 0.1 < \lambda_{H,cr}$. Initial conditions: $u_0 = v_0 = 0$, $\dot{u}_0 = 0.5$, $\dot{v}_0 = 0$

corresponds to large amplitude oscillations that obviously lead to high velocity and consequently accelerated energy dissipation. Then, the amplitude decreases and the system's slow flow approaches the SIM left stable branch after a rapid transition through the unstable branch. This slow buildup of energy and fast decay were referred to as SMR, in which the averaged flow performs relaxation oscillations as a result of the strong amplitude modulations caused by the slow-flow jump between both branches.

5 Qualitative comparison with computational and experimental results

As mentioned in the introduction, various experimental and computational studies were performed in the

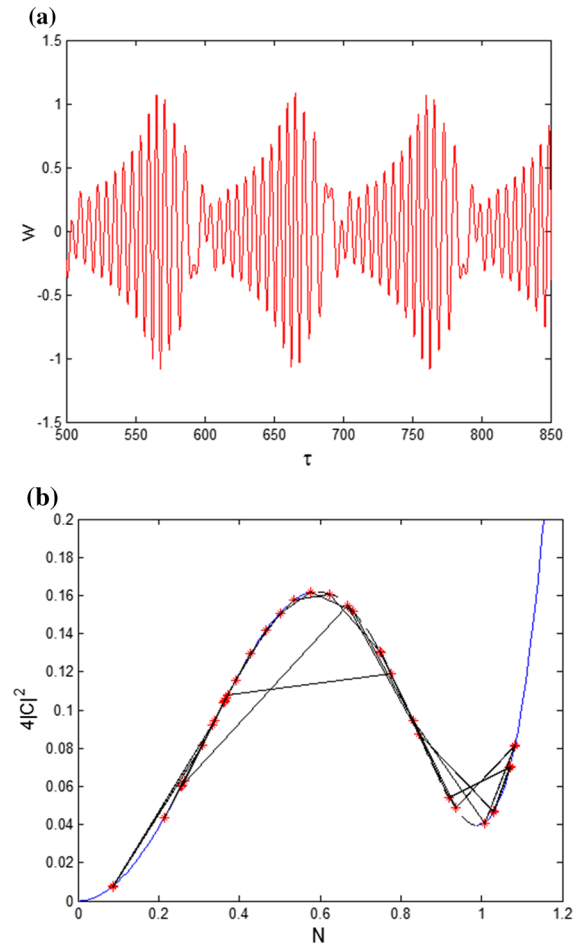


Fig. 13 SMR regime: $k = 4/3$, $\sigma = 5$, $\varepsilon = 0.05$, $\beta = 5$, $A = 0.3$, $\lambda = 0.2 \Rightarrow A > A_{SMR,cr} = 0.176$. Initial conditions: $u_0 = 0.29v_0 = -0.5$, $\dot{u}_0 = 0.9\dot{v}_0 = -0.15$

field of weakly nonlinear response regimes analysis of liquid sloshing in partially filled tank subjected to horizontal periodic ground excitation. Vast majority of the sloshing regimes documented are two dimensional, i.e., single periodic solutions, existence of two periodic solutions (amplitude jump), weakly nonlinear beatings, and the SMR. We present a ROM for partially filled liquid vessel exposed to horizontal ground excitation in Sect. 2. The response regimes are indicated, described and analyzed with the help of analytical and asymptotical tools in Sect. 3. Finally, the analytical predictions are validated numerically in Sect. 4. In this section, the dynamical regimes revealed in previous sections are compared qualitatively with several experimental and computational studies.

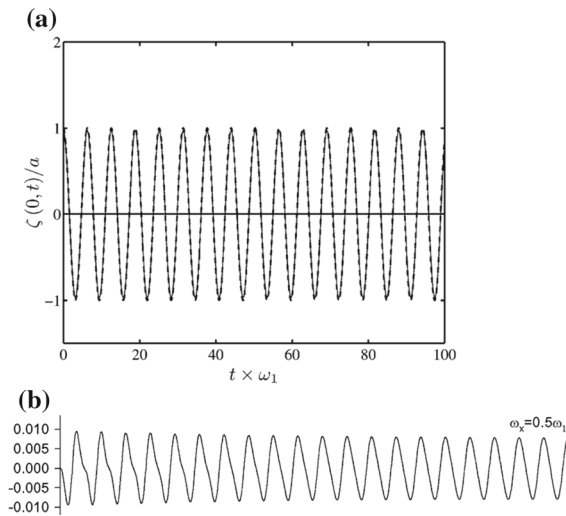


Fig. 14 Periodic sloshing responses in horizontally excited partially filled tanks; **a** Frandsen [9], normalized free-surface elevation at the left wall in fixed tank versus normalized time-scale, numerical solution (solid line) and second-order approximate solution (dashed line); **b** Chen and Nokes [68], time history of surface displacement at left tank wall

Periodic sloshing of single amplitude in horizontally excited liquid tanks is well studied and documented. This regime typically arises sufficiently far from the peak of the response curve, where jump phenomenon takes place due to softening or hardening nonlinearity (depends in liquid depth). Two examples are shown in Fig. 16.

Comparison between computational simulation and second-order approximate solution made by Frandsen [9] is presented in Fig. 14a. In this work, a fully nonlinear finite-difference model has been developed based on inviscid flow equations. The comparison shows good agreement for small horizontal forcing amplitude in comparison with second-order small perturbation theory. For large amplitude horizontal forcing, nonlinear effects are revealed both by the third-order single modal solution and by the fully nonlinear numerical model. The numerical simulation made by Chen and Nokes [68] shown in Fig. 16b presents periodic sloshing response due to horizontal excitation of half the frequency of the first sloshing mode, using independent finite-difference method. Multiple periodic solution associated with jump phenomenon near primary resonance was well documented in various studies [5, 8, 56, 69]. Hence, the multiple solution regime analyzed in Sect. 3.1.1 is fairly straightforward.

Hill and Frandsen [20] solved analytically the third-order amplitude evolution equation in terms of Jacobian elliptic functions to describe the evolution of weakly nonlinear sloshing wave, under assumptions of finite fluid depth and non-breaking waves. Classic hard-spring and soft-spring Duffing-type responses were revealed. Hill [19] performed weakly nonlinear analysis of the transient evolution of two-dimensional, standing waves in a rectangular basin using multiple-scales approach in vicinity of the critical fluid depth. The effects of detuning with respect to primary resonance, viscous damping, and cubic nonlinearity are considered. Both hardening and softening frequency response curves were produced, and beating responses were indicated.

Based on multidimensional modal analysis, Faltinsen [8] investigated numerically the influence of free liquid surface initial conditions. He showed that both the calculations and the experimental results indicate that both weakly and strongly modulated nonlinear transient waves take place. The maximum liquid elevation on the tank inner wall was almost twice larger than the linear model predictions. In addition, it was shown that the nonlinear effects become more dominant with decreasing liquid level and increasing forcing amplitude due to amplification of higher modes and failure of the single-dominant modal analysis.

Chosen results of the studies mentioned above are shown in Fig. 15a–c. Those results exhibit weakly quasiperiodic responses. The same sloshing regime was obtained from the current equivalent mechanical model, as shown in Fig. 15d. One can learn that there is a good qualitative agreement between the different results.

In the work by Zhang et al. [10], boundary element method (BEM) has been adopted to investigate the sloshing in 3D rectangular tanks in the vicinity of second-order resonance. Fully nonlinear free-surface boundary conditions were applied. Fluid free-surface elevation at the left tank wall is shown in Fig. 16a, with the corresponding spectrum distribution shown in Fig. 16b. Ikeda et al. [33] investigated nonlinear sloshing in rectangular tanks subjected to obliquely horizontal, harmonic excitation in vicinity of internal resonance condition 1:1 between the natural frequencies of predominate modes. Nonlinear modal equations of motion were derived, considering nine sloshing modes. Furthermore, Hopf bifurcation was documented, and amplitude modulated motions (referred to as AMMs),

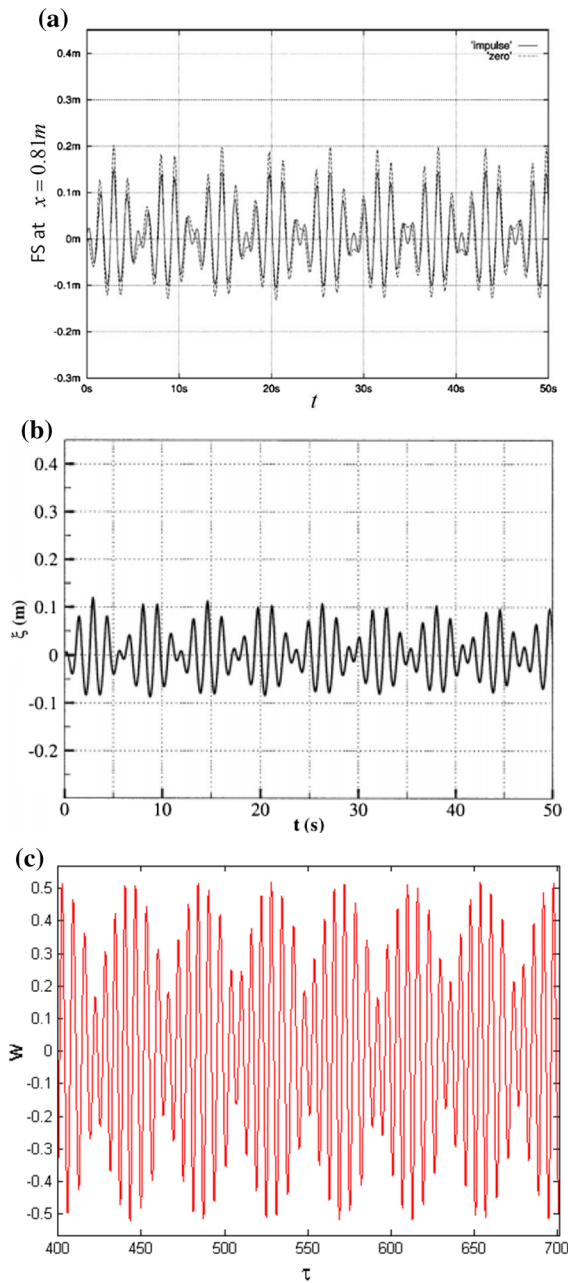


Fig. 15 Nonlinear beatings response regime in partially filled rectangular tank subjected to horizontal periodic excitation. Free-surface elevation at the left wall in horizontally excited tank; **a** Faltinsen [8]; **b** Hill [19]; **c** current model for parameter set: $\lambda = 0.4$, $A = 0.3$, $\sigma = -0.2$, $\varepsilon = 0.1$, $k = 5$

including chaotic motions, may appear depending on the value of the excitation frequency.

One can see that in the computational results shown in Fig. 16a, c the response amplitude envelope, i.e., the slow-flow dynamics, performs slow increase and fast decrease, alternately. The described regime has similar nature as SMR, analyzed in Sect. 3.2.2.

Regarding the response spectra shown in Fig. 16b, e, a, good qualitative agreement was obtained in terms of number of dominant frequencies. However, due to differences between the systems, those frequencies are related to different terms in the system's equations; in Fig. 16b, the lower frequency is related to the external excitation, and the higher frequency is related to the first sloshing mode. On the other hand, in Fig. 16e, the lower frequency is related to the external excitation, and the higher frequency—to the oscillations of the structure (DOF X, to be exact).

One can see in Figs. 14, 15 and 16 that there is a considerable qualitative agreement between the experimental and numerical results reported in the literature and the sloshing regimes described by the current ROM. This fact holds both for periodic responses, weak quasiperiodic responses and the SMRs.

6 Concluding remarks

A novel and relatively simple design of equivalent mechanical model with two degrees of freedom was suggested for qualitative description and understanding of the nonlinear two-dimensional sloshing regimes and their underlying dynamical mechanisms. The basic model may be reduced to the well-studied cubic NES with addition of the linear spring; this analogy yields efficient methods for assessment of non-planar and weakly nonlinear sloshing regimes inside a partially filled vessel. The analytical predictions were validated numerically. Finally, qualitative comparison demonstrates a similarity between the regimes detected by this model and the regimes documented experimentally and numerically.

It would be, of course, Naïve to expect good quantitative predictions for that simple model. Nevertheless, one can identify the basic response patterns predicted by the model (steady-state, weak modulation and the SMR) in the results supplied by more realistic analysis. This result, of course, leaves many questions. First, one should mention that the current model seems irrelevant

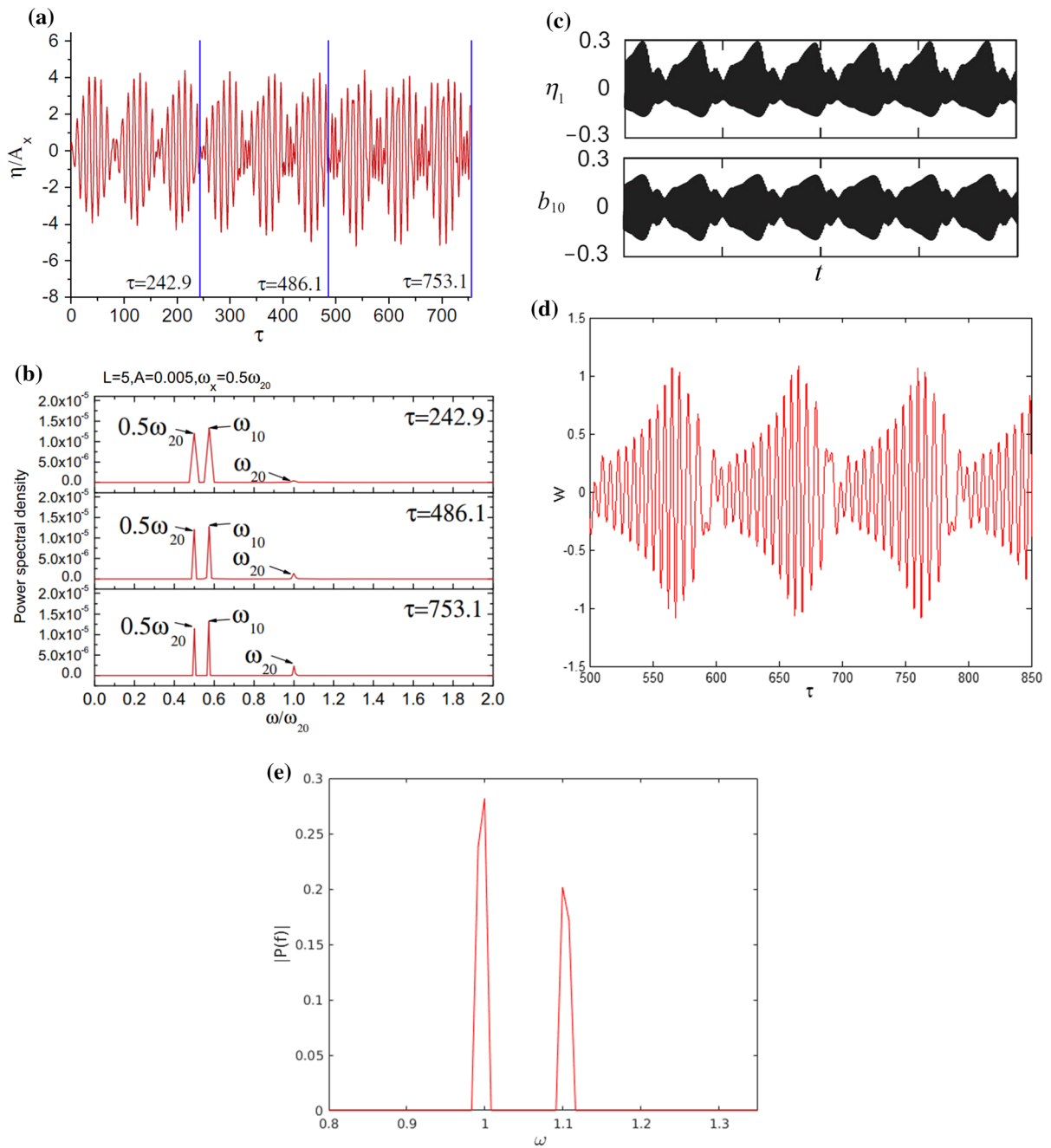


Fig. 16 Strongly modulated response regime (SMR); **a** Zhang et al. [10] normalized liquid elevation versus normalized time; **b** the corresponding spectrum where ω_{10} , ω_{20} are the natural frequencies of the first and second sloshing modes, and ω_x is the frequency of the external horizontal excitation; **c** Ikeda et al.

[33], liquid elevation η_1 and the first modal amplitude b_{10} versus normalized time; **d–e** current model, sloshing amplitude w for parameter set: $\lambda = 0.2$, $A = 0.3$, $\beta = 1$, $\sigma = 0$, $\varepsilon = 0.05$, $k = 4/3$

for three-dimensional sloshing regimes (e.g., swirling). Besides, it would be interesting to verify whether the model of the same dimensionality, but with different choice of the model potentials would be able to yield better predictions, or one should necessarily use the equivalent models with higher dimensionality. These issues will become a subject of further exploration..

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Appendix A

$$a_1 = \frac{(1 + \varepsilon) \lambda}{2} \quad (\text{A1})$$

$$a_2 = \frac{1}{2} (1 - \varepsilon \eta) - \frac{(1 + \varepsilon \sigma)^2}{2 (1 + \varepsilon) (1 - \sigma)} \quad (\text{A2})$$

$$a_3 = \frac{-3k (1 + \varepsilon)}{8} \quad (\text{A3})$$

$$a_4 = \frac{-A (1 + \varepsilon)}{2 (1 - \sigma)} \quad (\text{A4})$$

Appendix B

$$\alpha_1 = \lambda^2 + \frac{\sigma^2}{(1 - \sigma)^2} + \frac{2\beta\varepsilon\sigma}{1 - \sigma} + \beta^2\varepsilon^2 \quad (\text{B1})$$

$$\alpha_2 = \frac{3k}{2 (1 - \sigma)} (\sigma + \varepsilon\beta (1 - \sigma)) \quad (\text{B2})$$

$$\alpha_3 = \frac{9k^2}{16} \quad (\text{B3})$$

$$\alpha_4 = -\frac{A^2}{(1 - \sigma)^2} \quad (\text{B4})$$

Appendix C

$$c_{11} = \frac{i\varepsilon (1 - \sigma)}{2 (1 + \varepsilon)} \quad (\text{C1})$$

$$c_{12} = \frac{i\varepsilon (1 + \varepsilon\sigma)}{2 (1 + \varepsilon)} \quad (\text{C2})$$

$$c_{21} = \frac{i (1 + \varepsilon\sigma)}{2 (1 + \varepsilon)} \quad (\text{C3})$$

$$c_{22} = \frac{\lambda (1 + \varepsilon)}{2} \quad (\text{C4})$$

$$c_{23} = \frac{i}{2} \left(\frac{(1 - \varepsilon^2\sigma)}{(1 + \varepsilon)} - \varepsilon\beta (1 + \varepsilon) \right) \quad (\text{C5})$$

$$c_{24} = \frac{3ki (1 + \varepsilon)}{8} \quad (\text{C6})$$

Appendix D

$$\gamma_1 = \lambda (1 + \varepsilon) \quad (\text{D1})$$

$$\begin{aligned} \gamma_2 = & \frac{27}{64} k^2 (1 + \varepsilon)^2 N^4 \\ & + \frac{3k (1 + \varepsilon)}{4} \left(\frac{\varepsilon\sigma^2 - 1}{1 + \varepsilon} + \varepsilon\beta (1 + \varepsilon) \right) N^2 \\ & + \frac{1}{4} \left(\lambda^2 (1 + \varepsilon)^2 + (\varepsilon^2\sigma + 1) \right. \\ & \left. + 2\varepsilon\beta (\varepsilon^2\sigma - 1) + \varepsilon^2\beta^2 (1 + \varepsilon)^2 \right) \end{aligned} \quad (\text{D2})$$

$$\gamma_3 = \frac{1}{4} \varepsilon \lambda (\varepsilon\sigma^2 + 1) \quad (\text{D3})$$

$$\begin{aligned} \gamma_4 = & \frac{27}{256} \varepsilon^2 k^2 (1 - \sigma)^2 N^4 \\ & + \frac{3k\varepsilon^2 (1 - \sigma)}{16} (\sigma + \varepsilon\beta (1 - \sigma)) N^2 \\ & + \frac{\varepsilon^2}{16} \left(\lambda^2 (1 - \sigma)^2 + \sigma^2 + 2\varepsilon\beta\sigma (1 - \sigma) \right. \\ & \left. + \varepsilon^2\beta^2 (1 - \sigma)^2 \right) \end{aligned} \quad (\text{D4})$$

Appendix E

$$\bar{c}_1 = c_{11} + c_{22} + c_{23} \quad (\text{E1})$$

$$\bar{c}_2 = -c_{24} \quad (\text{E2})$$

$$\bar{c}_3 = c_{12}c_{21} - c_{11} (c_{22} + c_{23}) \quad (\text{E3})$$

$$\bar{c}_4 = c_{11}c_{24} \quad (\text{E4})$$

$$\bar{c}_5 = c_{11} + c_{21} \quad (\text{E5})$$

Appendix F

$$\xi_{11} = A \left(1 - \frac{3k}{4} N^2 \right) \quad (\text{F1})$$

$$\xi_{12} = \lambda A \quad (\text{F2})$$

$$\xi_{21} = \lambda A \quad (\text{F3})$$

$$\xi_{22} = -A \left(1 - \frac{9k}{4} N^2 \right) \quad (\text{F4})$$

$$\rho_1 = \lambda N \quad (\text{F5})$$

$$\begin{aligned} \rho_2 = & \frac{27k^2}{16} N^5 - \frac{3k}{4} (1 - 4\sigma) N^3 \\ & - \left(\sigma - \lambda^2 (1 - \sigma) \right) N \end{aligned} \quad (\text{F6})$$

References

- Shangguan, W.B., Chen, Y., Wang, Q., Rakheja, S.: Simulation of a partly filled tank vehicle combination in TruckSim and tank design optimisation. *Int. J. Heavy Veh. Syst.* **23**, 264 (2016)
- Yu, D., Li, X., Liu, H., Dong, J.: Research on liquid sloshing model of partially-filled tank by nonlinear external excitation. *J. Vibroeng.* **17**, 3224–3236 (2015)
- Yu, D., Li, X., Liu, H., Ren, Y., Dong, J., Wang, L.: Theory and experiments on driving stability of tank trucks under dangerous working conditions. *J. Vibroeng.* **17**, 2521–2534 (2015)
- Cox, P.A., Bowles, E.B., Bass, R.L.: Evaluation of liquid dynamic loads in slack LNG cargo tanks. *SNAME Trans.* **88**, 103–126 (1980)
- Abramson, H.N.: *The Dynamic Behavior of Liquids in Moving Containers*. NASA SP-106. NASA Special Publication 106 (1966)
- Dodge, F.T.: Analytical representation of lateral sloshing by equivalent mechanical models. *Dyn. Behav. Liq. Mov. Contain.* NASA SP-106 (1966)
- Cooper, R.M.: Dynamics of liquids in moving containers. *ARS J.* **30**, 725–729 (1960)
- Faltinsen, O.M., Rognebakke, O.F., Lukovsky, I.A., Timokha, A.N.: Multidimensional modal analysis of nonlinear sloshing in a rectangular tank with finite water depth. *J. Fluid Mech.* **407**, 201–234 (2000)
- Frandsen, J.B.: Sloshing motions in excited tanks. *J. Comput. Phys.* **196**, 53–87 (2004)
- Zhang, C., Li, Y., Meng, Q.: Fully nonlinear analysis of second-order sloshing resonance in a three-dimensional tank. *Comput. Fluids* **116**, 88–104 (2015)
- Bauer, H.F.: Nonlinear mechanical model for the description of propellant sloshing. *AIAA J.* **4**, 1662–1668 (1966)
- Pilipchuk, V.N., Ibrahim, R.A.: The dynamics of a non-linear system simulating liquid sloshing impact in moving structures. *J. Sound Vib.* **205**, 593–615 (1997)
- El-Sayad, M.A., Hanna, S.N., Ibrahim, R.A.: Parametric excitation of nonlinear elastic systems involving hydrodynamic sloshing impact. *Nonlinear Dyn.* **18**, 25–50 (1999)
- Ibrahim, R.A.: Recent advances in vibro-impact dynamics and collision of ocean vessels. *J. Sound Vib.* **333**, 5900–5916 (2014)
- Farid, M., Gendelman, O.V.: Internal resonances and dynamic responses in equivalent mechanical model of partially liquid-filled vessel. *J. Sound Vib.* **379**, 191–212 (2016)
- Lepelletier, T.G., Raichlen, F.: Nonlinear oscillations in rectangular tanks. *J. Eng. Mech.* **114**, 1–23 (1988)
- Moiseev, N.N.: On the theory of nonlinear vibrations of a liquid of finite volume. *J. Appl. Math. Mech.* **22**, 860–872 (1958)
- Faltinsen, O.M., Rognebakke, O.F., Timokha, A.N.: Resonant three-dimensional nonlinear sloshing in a square-base basin. *J. Fluid Mech.* **487**, 1–42 (2003)
- Hill, D.F.: Transient and steady-state amplitudes of forced waves in rectangular basins. *Phys. Fluids* **15**, 1576–1587 (2003)
- Hill, D., Frandsen, J.: Transient evolution of weakly nonlinear sloshing waves: an analytical and numerical comparison. *J. Eng. Math.* **53**, 187–198 (2005)
- Stolbetsov, V.I.: On oscillations of a fluid in the tank having the shape of rectangular parallelepiped. *Mekh. Zhidk. Gaza*, 67–76 (in Russian) (1967)
- Narimanov, G.S.: Movement of a tank partly filled by a fluid: the taking into account of non-smallness of amplitude. *J. Appl. Math. Mech.* **21**, 513–524 (1957)
- Bryant, P.J., Stiassnie, M.: Different forms for nonlinear standing waves in deep water. *J. Fluid Mech.* **272**, 135 (1994)
- Waterhouse, D.D.: Resonant oscillations of gases and liquids in three dimensions. PhD thesis, Oxford University (1995)
- Ockendon, H., Ockendon, J.R., Waterhouse, D.D.: Resonant three-dimensional nonlinear sloshing in a square-base basin. *J. Fluid Mech.* **315**, 317 (1996)
- Faltinsen, O.M., Rognebakke, O.F., Timokha, A.N.: Resonant three-dimensional nonlinear sloshing in a square-base basin. Part 2. Effect of higher modes. *J. Fluid Mech.* **523**, 199–218 (2005)
- Faltinsen, O.M., Rognebakke, O.F., Timokha, A.N.: Transient and steady-state amplitudes of resonant three-dimensional sloshing in a square base tank with a finite fluid depth. *Phys. Fluids* **18**, 012103 (2006)
- Faltinsen, O.M., Timokha, A.N.: *Sloshing*. Cambridge University Press, Cambridge (2009)
- Forbes, L.K.: Sloshing of an ideal fluid in a horizontally forced rectangular tank. *J. Eng. Math.* **66**, 395–412 (2010)
- Chester, W.: Resonant oscillations of water waves. I. Theory. *Proc. R. Soc. A Math. Phys. Eng. Sci.* **306**, 5–22 (1968)
- Chester, W., Bones, J.A.: Resonant oscillations of water waves. II. Experiment. *Proc. R. Soc. A Math. Phys. Eng. Sci.* **306**, 23–39 (1968)
- Ikeda, T., Ibrahim, R.A., Harata, Y., Kuriyama, T.: Nonlinear liquid sloshing in a square tank subjected to obliquely horizontal excitation. *J. Fluid Mech.* **700**, 304–328 (2012)
- Ikeda, T., Harata, Y., Osasa, T.: Internal resonance of nonlinear sloshing in rectangular liquid tanks subjected to obliquely horizontal excitation. *J. Sound Vib.* **361**, 210–225 (2016)
- Welch, J.E., Harlow, F.H., Shannon, J.P., Daly, B.J.: The MAC method—a computing technique for solving viscous. Incompressible, transient fluid-flow problems involving free surfaces. Los Alamos Scientific Laboratory Report No. 3425 (1966)
- Hughes, T.J.R., Liu, W.K., Zimmermann, T.K.: Lagrangian–Eulerian finite element formulation for incompressible viscous flows. *Comput. Methods Appl. Mech. Eng.* **29**, 329–349 (1981)
- Nakayama, T., Washizu, K.: The boundary element method applied to the analysis of two-dimensional nonlinear sloshing problems. *Int. J. Numer. Methods Eng.* **17**, 1631–1646 (1981)
- Lin, P., Liu, P.L.F.: A numerical study of breaking waves in the surf zone. *J. Fluid Mech.* **359**, 239–264 (1998)
- Faltinsen, O.M.: A numerical nonlinear method of sloshing in tanks with two-dimensional flow. *J. Ship Res.* **22**, 193–202 (1978)
- Zhang, C.: Application of an improved semi-Lagrangian procedure to fully-nonlinear simulation of sloshing in non-wall-sided tanks. *Appl. Ocean Res.* **51**, 74–92 (2015)

40. Graham, E.W., Rodriquez, A.M.: The characteristics of fuel motion which affect air-plane dynamics. *J. Appl. Mech.* **19**(3), 381–388 (1952)
41. Ibrahim, R.A., Pilipchuk, V.N., Ikeda, T.: Recent advances in liquid sloshing dynamics. *Appl. Mech. Rev.* **54**, 133 (2001)
42. Farid, M., Gendelman, O.V.: Tuned pendulum as nonlinear energy sink for broad energy range. *J. Vib. Control* **23**, 373–388 (2015)
43. Kana, D.D.: A model for nonlinear rotary slosh in propellant tank. *J. Spacecr. Rockets* **24**, 169–177 (1987)
44. Farid, M., Gendelman, O.V.: Response regimes in equivalent mechanical model of strongly nonlinear liquid sloshing. *Int. J. Non-Linear Mech.* **94**, 146–159 (2017)
45. Farid, M., Levy, N., Gendelman, O.V.: Vibration mitigation in partially liquid-filled vessel using passive energy absorbers. *J. Sound Vib.* **406**, 51–73 (2016)
46. Gendelman, O.V., Starosvetsky, Y.: Quasi-periodic response regimes of linear oscillator coupled to nonlinear energy sink under periodic forcing. *J. Appl. Mech.* **74**, 325 (2007)
47. Gendelman, O.V., Starosvetsky, Y., Feldman, M.: Attractors of harmonically forced linear oscillator with attached nonlinear energy sink I: description of response regimes. *Nonlinear Dyn.* **51**, 31–46 (2007)
48. Starosvetsky, Y., Gendelman, O.V.: Strongly modulated response in forced 2DOF oscillatory system with essential mass and potential asymmetry. *Phys. D Nonlinear Phenom.* **237**, 1719–1733 (2008)
49. Starosvetsky, Y., Gendelman, O.V.: Response regimes of linear oscillator coupled to nonlinear energy sink with harmonic forcing and frequency detuning. *J. Sound Vib.* **315**, 746–765 (2008)
50. Graham, E.W.: The Forces Produced by Fuel Oscillation in a Rectangular Tank. Douglas Aircraft Company, Incorporated, Santa Monica (1951)
51. Partom, I.S.: Numerical calculation of equivalent moment of inertia for a fluid in a cylindrical container with partitions. *Int. J. Numer. Methods Fluids* **5**, 25–42 (1985)
52. Warner, R.W., Caldwell, J.T.: Experimental evaluation of analytical models for the inertias and natural frequencies of fuel sloshing in circular cylindrical tanks. National Aeronautics and Space Administration (1961)
53. Daqaq, M.F., Xu, Y., Lacarbonara, W.: Nonlinearity of finite-amplitude sloshing in rectangular containers. *J. Appl. Mech.* **84**, 031002 (2016)
54. Aliabadi, S., Johnson, A., Abedi, J.: Comparison of finite element and pendulum models for simulation of sloshing. *Comput. Fluids* **32**, 535–545 (2003)
55. Malhotra, P.K., Wenk, T., Wieland, M.: Simple procedure for seismic analysis of liquid-storage tanks. *Struct. Eng. Int.* **10**, 197–201 (2000)
56. Fultz, D.: An experimental note on finite-amplitude standing gravity waves. *J. Fluid Mech.* **13**, 193 (1962)
57. Hermann, M., Timokha, A.: Modal modelling of the nonlinear resonant fluid sloshing in a rectangular tank II: secondary resonance. *Math. Model. Methods Appl. Sci.* **18**, 1845–1867 (2008)
58. Gu, X.M., Sethna, P.R.: Resonant surface waves and chaotic phenomena. *J. Fluid Mech.* **183**, 543 (1987)
59. Gu, X.M., Sethna, P.R., Narain, A.: On three-dimensional nonlinear subharmonic resonant surface waves in a fluid: part I—theory. *J. Appl. Mech.* **55**, 213 (1988)
60. Babitsky, V.I.: Theory of vibro-impact systems and applications. Springer (2013)
61. Gendelman, O.V.: Analytic treatment of a system with a vibro-impact nonlinear energy sink. *J. Sound Vib.* **331**, 4599–4608 (2012)
62. Gendelman, O.V., Alloni, A.: Dynamics of forced system with vibro-impact energy sink. *J. Sound Vib.* **358**, 301–314 (2015)
63. Goldstein, H.: Classical Mechanics. Pearson Education India, Bangalore (2011)
64. Starosvetsky, Y., Gendelman, O.V.: Vibration absorption in systems with a nonlinear energy sink: nonlinear damping. *J. Sound Vib.* **324**, 916–939 (2009)
65. Manevitch, L.I.: Complex representation of dynamics of coupled nonlinear oscillators. In: Uvarova, L.A., Arinstein, A.E., Latyshev, A.V. (eds.) *Mathematical Models of Non-linear Excitations, Transfer, Dynamics, and Control in Condensed Systems and Other Media*, pp. 269–300. Springer, Boston (1999)
66. Gendelman, O., Manevitch, L.I., Vakakis, A.F., M'closkey, R.: Energy pumping in nonlinear mechanical oscillators: part i: dynamics of the underlying Hamiltonian systems. *J. Appl. Mech.* **68**, 34–41 (2001)
67. Guckenheimer, J., Hoffman, K., Weckesser, W.: Bifurcations of relaxation oscillations near folded saddles. *Int. J. Bifurc. Chaos* **15**, 3411–3421 (2005)
68. Chen, B.-F., Nokes, R.: Time-independent finite difference analysis of fully non-linear and viscous fluid sloshing in a rectangular tank. *J. Comput. Phys.* **209**, 47–81 (2005)
69. Ibrahim, R.A., Li, W.: Parametric and autoparametric vibrations of an elevated water tower, part II: autoparametric response. *J. Sound Vib.* **121**, 429–444 (1988)